Knowledge-Infused Temporal Fusion Transformer for Interpretable Riverbank Deformation Forecasting Using Sentinel-1 InSAR Time Series

Weijie Ding, Heqin Cheng, Zhaocai Wang, Dongping Chen, Junhao Wu, Ruonan Bao, Wenqiu Qu, and Irene Cheng

***Abstract*—Forecasting riverbank deformation requires characterizing complex spatiotemporal heterogeneity governed by coupled hydro-meteorological forcing and site-specific geomorphology. Although deep learning has shown promise for geohazard prediction, existing models for fluvial instability largely operate as black boxes, lacking both physical transparency and spatial transferability. This study proposes the knowledge-infused variational mode decomposition (VMD)-temporal fusion Transformer (TFT), knowledge-infused VMD-TFT (KI-VTFT), a physically constrained and interpretable framework that embeds domain knowledge into data-driven forecasting through two complementary pathways: 1) signal-level priors, injected via Bayesian-optimized VMD (BO-VMD), which partitions displacement into trend, periodic, and residual components; and 2) spatial-level priors, encoded through seven static geomorphic attributes that condition predictions on terrain-controlled susceptibility. The framework was applied to continuous displacement fields derived from 177 Sentinel-1 images (2021–2024) using small baseline subset interferometric synthetic aperture radar (SBAS-InSAR) across three subreaches of the Lower Yangtze River (LYR). At the five-step (60-day) forecasting horizon, KI-VTFT achieved a root-mean-square error (RMSE) of 2.29 mm and an R^2 of 0.97, reducing RMSE by 25.96%–65.12% relative to gated recurrent units (GRUs), long short-term memory (LSTM), temporal convolutional networks (TCNs), and Transformer baselines. Interpretability analysis revealed that VMD components drove 64.52% of encoder feature importance, while distance to thalweg (22.51%) and elevation (20.78%) dominated the static controls; temporal attention weights peaked during the May–August flood season, consistent with regional hydrological cycles. Independent validation using uncrewed aerial vehicle and uncrewed surface vessel (UAV–USV) bathymetric surveys, acoustic Doppler current profiler (ADCP) flow measurements, and sediment sampling supported the physical plausibility of model-identified drivers. Furthermore, zero-shot transfer to the separate Wuwei Reach maintained an RMSE of 2.77 mm without retraining, outperforming the strongest baseline, the Transformer (3.65 mm), under identical conditions and demonstrating the framework's potential for cross-reach generalization.**

***Index Terms*—InSAR time series, interpretable deep learning, riverbank deformation forecasting, spatial transferability, temporal fusion Transformer (TFT).**

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I. Introduction

RIVERBANK collapse constitutes a critical fluvial geohazard in alluvial systems, compromising embankment integrity, navigation safety, and regional socioeconomic stability [1], [2]. The Lower Yangtze River (LYR), extending 938 km from Hukou to the estuary, has experienced persistent channel degradation since the 2003 impoundment of the Three Gorges Dam, as sediment trapping by upstream reservoir cascades intensified clear-water scour and destabilized bank slopes [3], [4]. The escalating magnitude of bank instability across this critical economic corridor necessitates reliable deformation monitoring and predictive capabilities to support proactive risk management.

Spaceborne interferometric synthetic aperture radar (InSAR), particularly the small baseline subset InSAR (SBAS-InSAR) technique, enables wide-area deformation retrieval with millimeter-level accuracy while mitigating spatiotemporal decorrelation in vegetated riparian zones [5], [6], [7]. The Sentinel-1 archive provides deformation time series at a 12-day revisit interval, yielding the dense temporal sampling essential for characterizing progressive bank displacement [8]. While such monitoring yields high-quality historical records, operational early warning demands the ability to forecast future displacement, thereby transforming retrospective observations into prospective risk assessments.

Forecasting approaches for geohazard displacement have evolved rapidly. Traditional statistical methods (gray system theory, regression) assume quasi-linear behavior and struggle to capture nonlinear dynamics [9], [10], [11]. Machine

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learning techniques such as support vector regression (SVR) and extreme learning machines (ELMs) improved accuracy but treat forecasting as a static mapping without encoding temporal dependencies [12], [13], [14]. Recurrent architectures, specifically long short-term memory (LSTM) and gated recurrent units (GRUs), explicitly model sequential dependencies [15], [16], [17], [18]. However, the iterative nature of RNNs can limit scalability for longer sequences. Temporal convolutional networks (TCNs) enable parallel computation via dilated causal convolutions [19], while the Transformer architecture models global dependencies through self-attention [20]. The temporal fusion Transformer (TFT) further integrates variable selection networks and interpretable multihead attention, quantifying the contribution of individual driving factors and historical time lags [21], [22], [23]. Concurrently, signal decomposition techniques, particularly variational mode decomposition (VMD), have been coupled with deep learning to separate trend, periodic, and stochastic components prior to sequence modeling, significantly improving forecast accuracy in geohazard applications [24], [25], [26]. Deep learning prediction driven directly by InSAR-derived displacement has likewise advanced, coupling variational-mode-decomposed Sentinel-1 time series with spatiotemporal kinematic feature learning for slope-scale forecasting [27] and capturing the heterogeneous long-term evolution of InSAR-monitored slopes with spatiotemporal graph convolutional networks [28]. The paradigm of infusing domain knowledge into interpretable deep learning for deformation forecasting was established in the landslide domain by Ma and Mei [29], initially through the Transformer-based LFIT network and subsequently with explicit treatment of spatiotemporal correlations [30]. The descriptor knowledge-infused approach, widely used in the machine learning literature for architectures that embed domain priors, applies naturally to both lines of work. In the latter, slope displacements recorded by GNSS instruments are forecast by merging internal geotechnical attributes with external triggers as variable-level inputs, and local heterogeneity is handled by treating each instrumented point as an individual identifier.

Critically, these methodological advances have been developed and validated almost exclusively for gravitational landslides in mountainous terrain. Riverbank instability is governed by fundamentally different mechanisms, including hydro-mechanical coupling, cyclic water-level fluctuations, and subaqueous scour–fill dynamics, which distinguish riparian deformation from hillslope failure [31], [32]. Systematic application of deep sequence models to InSAR-derived riverbank displacement remains scarce. Despite the progress outlined above, four critical limitations persist.

1) The decomposition-based prediction paradigm has not yet been systematically investigated for InSAR-observed riverbank deformation. Riverbank deformation is a superposition of long-term erosional trends, seasonal hydro-meteorological oscillations, and stochastic noise; treating these as a composite input forces models to implicitly disentangle deterministic dynamics from random disturbances, degrading forecast reliability [33].
2) Conventional models treat monitoring points as independent samples, neglecting spatial heterogeneity driven by site-specific geomorphology. Failure to encode terrain attributes (e.g., elevation, slope, curvature, and thalweg proximity) as conditional covariates prevents differentiation of susceptibility across heterogeneous bank environments, limiting transferability to unsurveyed locations [34].
3) Compounding these input- and spatial-level deficiencies, existing interpretability analyses in geohazard forecasting predominantly identify which factors matter [35], yet operational riverbank early warning additionally requires understanding when these factors exert influence and whether model-identified drivers are physically consistent with observed geomorphic processes. This represents a gap in joint temporal attribution and physical verification that remains largely unaddressed.
4) The physical plausibility of deep-learning-based deformation predictions has seldom been corroborated through independent field observations [36], [37]. In fluvial settings, where deformation is coupled to subaqueous morphodynamics invisible to satellite observation, reliance on statistical validation alone is particularly problematic.

To address these limitations, we propose the knowledge-infused VMD-TFT (KI-VTFT), a framework that integrates physical priors into data-driven forecasting via two complementary pathways: signal-level knowledge injection through Bayesian-optimized VMD (BO-VMD), which decomposes displacement into trend, periodic, and residual components; and spatial-level knowledge encoding through static geomorphic attributes as time-invariant TFT covariates. Relative to the knowledge-infused landslide models of Ma and Mei [29], [30], the present framework departs in more than just the shift from GNSS-monitored hillslopes to InSAR-observed riverbanks. The priors act on signal structure and terrain rather than entering as additional input variables, and the conditioning relies on transferable terrain attributes rather than point-specific identifiers, which keeps the trained model applicable at locations never seen during training. The framework is further subjected to closed-loop validation against independent multiplatform subaqueous field observations and to zero-shot evaluation on a separate reach, two elements that have not, to our knowledge, been reported in that line. The principal contributions are fourfold.

1) We adapt an established decomposition-based prediction paradigm to InSAR-observed riverbank deformation, jointly optimizing the VMD parameters K and α with the hyperparameters of the downstream TFT, and couple the resulting signal-level prior, which resolves displacement into trend, periodic, and residual components, with a spatial-level prior from seven static geomorphic attributes encoding terrain-controlled susceptibility.
2) We exploit the variable selection networks and the shared-value multihead attention native to the TFT architecture [21] to obtain feature importance rankings and critical historical time windows, and we verify these

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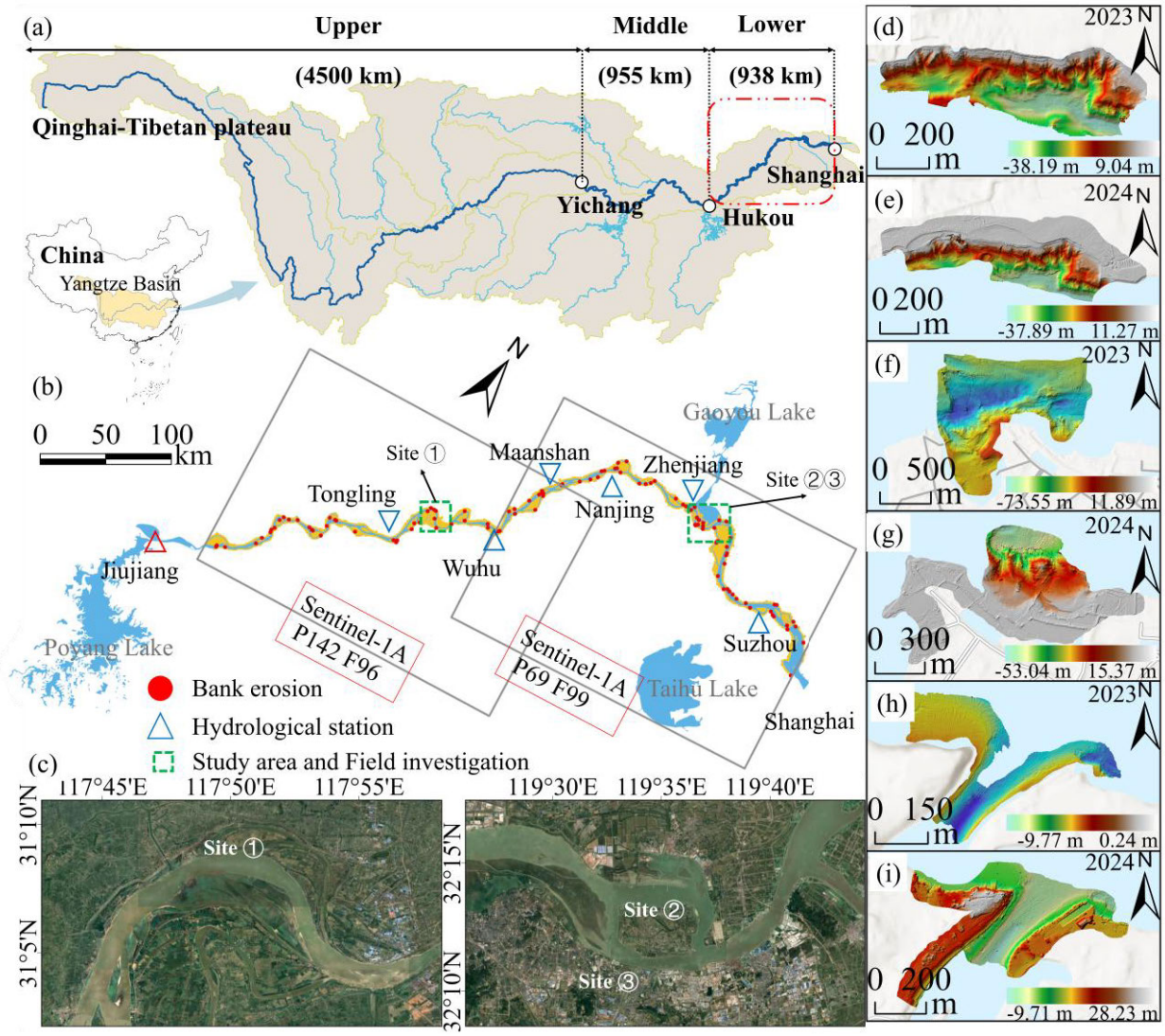


Fig. 1. Geographic location and geomorphic setting of the study area. (a) Yangtze River Basin in China, with the longitudinal division into the upper, middle, and lower reaches; the red dashed box marks the LYR. (b) LYR from Hukou to the estuary, showing the Sentinel-1A footprints (P142 F96 and P69 F99), historical bank-erosion sites, hydrological stations, and the three field-investigation sites (green dashed boxes). (c) High-resolution optical imagery of the survey sites with geographic coordinates: site ① near Tongling (left) and sites ② and ③ near Zhenjiang (right). (d)–(i) Multitemporal UAV–USV land–water DEMs (0.5-m spatial resolution) acquired in August 2023 and August 2024: site ① in (d) and (e), site ② in (f) and (g), and site ③ in (h) and (i). Elevation values are in meters; color scales are shown alongside each panel.

attributions against hydrological records and geomorphic field evidence, connecting data-driven inference to process understanding.

3) We evaluate zero-shot cross-reach transfer by applying the model trained on one reach directly to an independent reach without fine-tuning or retraining, testing whether the embedded priors generalize under analogous hydro-geomorphic settings rather than fitting site-specific artifacts.
4) We corroborate the physical plausibility of predicted deformation patterns through stable-area validation, geomorphic cross-referencing, and independent multiplatform field observations (repeat uncrewed aerial vehicle and uncrewed surface vessel (UAV–USV) bathymetry, acoustic Doppler current profiler (ADCP) flow profiling, sediment sampling), closing a verification gap between AI-based inference and process-based geomorphic understanding.

II. Study Area and Data Acquisition

A. Study Area

The LYR extends approximately 938 km from Hukou to the estuary, traversing Anhui, Jiangsu, and Shanghai as one of China's most critical economic corridors (Fig. 1). Geomorphologically, this reach exhibits a characteristic node-and-internode planform, defined by alternating constricted (single-channel) and bifurcated segments. Subject to the systemic clear-water scour regime induced by upstream sediment trapping described previously, these transitional river segments have become hotspots for riverbank instability due to their sensitive hydro-geomorphic boundary conditions [38].

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Within this regional context, we selected two representative high-risk subreaches for detailed investigation: 1) Wuwei Reach in Anhui Province and 2) Zhenyang Reach in Jiangsu Province, which encompasses Hechangzhou Island and the Nanyunhe channel [Fig. 1(b) and (c)]. Both subreaches feature typical goose-head bifurcated channel patterns. Hydrodynamically, the superposition of bend-induced secondary circulation and high-velocity thalweg flow intensifies near-bank scour. Geotechnically, the banks possess a binary stratigraphic structure with low scour resistance, comprising an upper cohesive layer (silty clay or loam) overlying a lower noncohesive stratum of fine-to-medium sand [39]. This configuration is highly susceptible to basal undermining during high-flow events, leading to mass failure. Consequently, the frequent channel migration and distinct scour–fill dynamics characteristic of these reaches serve as a rigorous testbed for evaluating the KI-VTFT framework.

B. Space–Air–Ground–Water Integrated Data Acquisition

1) SBAS-InSAR Processing and Deformation Retrieval: To retrieve reliable surface deformation in vegetated fluvial environments, we employed the small baseline subset (SBAS) technique, which effectively mitigates spatiotemporal decorrelation by constructing interferometric networks with short baselines [40]. The dataset comprised 177 C-band ascending Sentinel-1 single look complex (SLC) images acquired between November 2021 and December 2024, covering Wuwei and Zhenyang subreaches. Interferometric processing was conducted using ENVI SARscape 5.6.

Topographic phase removal was performed using the Diluvium digital elevation model (DEM) as the primary reference [41]. This 30-m global coastal DEM, derived by LightGBM-based correction of the Copernicus DEM, attains higher vertical accuracy than SRTM, FABDEM, and CoastalDEM over low-lying floodplains, making it well suited to the low-relief banks of the Lower Yangtze. Over the steep near-bank transition zones, however, this 30-m global product represents the abrupt local terrain only approximately, and the resulting elevation error is the dominant source of residual topographic phase in the deformation estimate. To suppress this residual, the reference DEM was locally refined with the 0.5-m integrated UAV–USV land–water surface within the survey footprints. The high-resolution survey was first aggregated to the Diluvium DEM grid to provide accurate near-bank cell elevations; in the overlapping near-bank zone, the two elevation sources were combined by weighted averaging in which the survey-derived elevations dominated within the survey footprint and transitioned smoothly to the regional DEM at its margins. The refined DEM was then used to simulate and subtract the topographic phase during differential interferogram generation, reducing near-bank topographic residuals prior to phase unwrapping. The interferometric network was constructed using perpendicular and temporal baseline thresholds of 2% of the critical baseline and 90 days, respectively, yielding 618 pairs. To suppress phase noise, interferograms underwent 4×1 (range $\times$ azimuth) multilooking and Goldstein adaptive filtering.

Atmospheric phase screen (APS) artifacts were corrected using the Generic Atmospheric Correction Online Service (GACOS), which integrates ERA5 reanalysis and GNSS observations to estimate high-resolution tropospheric delays. Phase unwrapping was executed via the minimum cost flow (MCF) algorithm with a coherence threshold of 0.6 [42]. Finally, displacement time series were retrieved to characterize the spatiotemporal evolution of riverbank instability. Line-of-sight (LOS) displacements were then converted to vertical displacement through division by the cosine of the local incidence angle, which is close to 40° for both tracks, following established practice in single-geometry monitoring of low-relief coastal and fluvial terrain [7]. The projection reflects the acquisition geometry rather than an assumption of purely vertical bank motion. At this geometry, the LOS observation weights vertical motion by about 0.77 and east–west motion by about 0.62, whereas north–south motion enters with a weight below 0.15. Because the monitored bank segments front channel sections oriented broadly east to west (Fig. 1), the bank-normal direction along which lateral failure develops coincides with this strongly attenuated axis, so lateral motion contributes little to the LOS record and does not appreciably bias the converted vertical series. The coherent scatterers available to InSAR occupy the subaerial bank top, where toe scour, deposition, and consolidation are expressed as settlement and uplift. The converted series thus capture the vertical expression of bank deformation, the component independently corroborated by the repeat bathymetric and field observations (Section V-B).

The reference point was automatically selected on a built-up structure (32.17°N, 119.57°E) with a temporal coherence of 0.85. The interferometric networks comprised 283 pairs for Zhenyang Reach and 335 pairs for Wuwei Reach, totaling 618 pairs (Fig. S1), constructed with the perpendicular and temporal baseline thresholds described above. Pixels with temporal coherence below 0.6 were excluded from the final displacement products.

To characterize the magnitude of the applied correction, we analyzed the estimated atmospheric phase for all 82 acquisition epochs of Zhenyang Reach. The per-epoch GACOS correction exhibited a spatial mean ranging from −5.13 to +4.14 mm, with an average spatial standard deviation of 3.6 mm. The effectiveness of the correction was then assessed using before-and-after statistics. Over the stable areas defined in Section IV-A2, GACOS correction lowered the per-epoch spatial standard deviation of the detrended displacement residuals for 80% of the epochs, by 47% on average, and reduced the semivariogram sill of the detrended velocity residuals by about 70% [Fig. S2(b) and (c)]. The residual against an independent leveling benchmark likewise decreased, from a root-mean-square error (RMSE) of 4.0–3.1 mm [Fig. S2(a)], as detailed in Section IV-A2.

2) UAV–USV Cooperative Land–Water Survey: To characterize micro-scale geomorphic evolution and validate InSAR-derived deformation, we conducted high-resolution integrated land–water topographic surveys at three key sites [Fig. 1(c)] between August 2023 and August 2024. Surveys employed a tidal-synchronized strategy: terrestrial and exposed-bank mapping was performed during low-tide windows using a

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UAV-based light detection and ranging (LiDAR) system, while underwater bathymetry was acquired during high-tide periods using a USV-mounted multibeam echosounder (MBES). This approach ensured seamless coverage across the critical subaerial–subaqueous transition zone [43].

UAV-LiDAR acquisition followed parallel flight paths with 80% along-track and 70% cross-track overlap, maintained at an altitude of 100 m and a speed of 5 m/s. The MBES operated at 400 kHz with a 120° swath width and equidistant beam spacing. Positioning for both platforms was corrected via GNSS real-time kinematic (RTK) networks, supplemented by static base stations in signal-shadowed areas to ensure geometric rigor. All data were referenced to China Geodetic Coordinate System 2000 (CGCS2000) and the 1985 National Elevation Datum, generating a fused DEM with a positional accuracy better than 5 cm.

To investigate the hydrodynamic and sedimentary drivers of bank instability, synchronous in situ measurements were acquired. A USV-mounted Teledyne RDI ADCP profiled cross-sectional flow velocity and direction. Concurrently, surface sediment samples were collected along the main channel via a box corer and analyzed for grain-size distribution using laser diffraction (Beckman Coulter LS13320). These datasets quantify local scour resistance and sediment transport dynamics.

3) Dynamic Covariate Construction: The rigorous selection of conditioning factors is critical for robust deep learning performance. Informed by established mechanisms, riverbank displacement is primarily driven by external hydrometeorological forcing (precipitation and water-level fluctuations) and modulated by antecedent deformation states and local geomorphic conditions [44]. Consequently, we constructed a comprehensive feature set integrating both time-varying (dynamic) covariates and time-invariant (static) attributes to input into the TFT.

Dynamic Covariates: To align with the 12-day revisit cycle of Sentinel-1, all dynamic variables were aggregated to commensurate temporal scales. Precipitation forcing was parameterized as cumulative rainfall over antecedent windows of 12, 24, and 36 days, capturing both immediate infiltration and soil moisture saturation lags. Daily precipitation records were acquired from the National Tibetan Plateau Data Center (https://data.tpdc.ac.cn/). Hydrodynamic forcing was characterized by two metrics computed over 12-day intervals: 1) mean water level, representing the overall saturation status; and 2) water-level fluctuation amplitude, representing the intensity of seepage pressure variations within the bank matrix. Water-level records were sourced from the Yangtze River Hydrological Yearbook. These five indicators collectively encapsulate the external triggers associated with seasonal hydrological variability.

4) Static Geomorphic Attribute Encoding: To encode spatial heterogeneity in terrain-controlled susceptibility, seven attributes were derived from a 30-m DEM generated by fusing the Diluvium DEM with high-resolution UAV–USV bathymetry: 1) elevation (gravitational potential); 2) slope (driving stress); 3) aspect (solar exposure and drying efficiency); 4) curvature (flow convergence/divergence); 5) distance to water (proximity to the erosion interface); 6) distance to thalweg (exposure to main-flow scour); and 7) distance to road (proxy for anthropogenic loading). These static covariates are embedded into the TFT encoder to provide the geomorphic context required to interpret site-specific deformation responses.

III. Methodology

A. Overview of the KI-VTFT Framework

This study proposes the KI-VTFT, an interpretable deep learning framework designed to forecast riverbank instability by systematically integrating physical priors with data-driven sequence modeling (Fig. 2). To enhance predictive capability and physical consistency, the framework synthesizes multisource observational data (SBAS-InSAR displacement, precipitation, and water levels) with two distinct categories of domain knowledge: 1) signal-level knowledge, injected via Bayesian optimization (BO)-VMD, which partitions raw nonstationary displacement series into three physically distinct components—trend, periodic, and residual—each supplied to the forecasting backbone as an independent input; and 2) spatial-level knowledge, encoded via static geomorphic attributes that characterize terrain-controlled susceptibility. By fusing these priors within a TFT architecture, the method not only improves forecasting accuracy but also quantifies the temporal contribution of specific drivers, thereby bridging the gap between black-box deep learning and mechanistic process understanding. The architecture is structured into five functional modules (Fig. 2).

Module ①) *Data Collection and Preprocessing:* Integrates multisource observations from the space–air–ground–water monitoring system to construct high-dimensional spatiotemporal datasets (Section II-B).

Module ②) *Spatial Knowledge Encoding:* Embeds static geomorphic attributes as time-invariant covariates, conditioning the model on site-specific susceptibility contexts (Section II-B4).

Module ③) *Signal Knowledge Injection:* Applies BO-VMD to decompose raw displacement sequences into three physically distinct components—trend, periodic, and residual—all of which are fed into the TFT encoder as independent observed inputs (Section III-B).

Module ④) *Spatiotemporal Forecasting:* Serves as the core predictive engine, where the TFT integrates decomposed signals and static metadata. It employs variable selection networks to identify critical drivers, LSTM encoders/decoders to capture temporal dynamics over a sliding window $(t - k, \ldots, t + \tau_{max})$, and an interpretable multihead attention mechanism to quantify the temporal weight of influence (Section III-C).

Module ⑤) *Physical Validation:* Establishes a feedback loop via independent field surveys (UAV, LiDAR, USV MBES, ADCP, and sediment), verifying the physical plausibility of both the predicted

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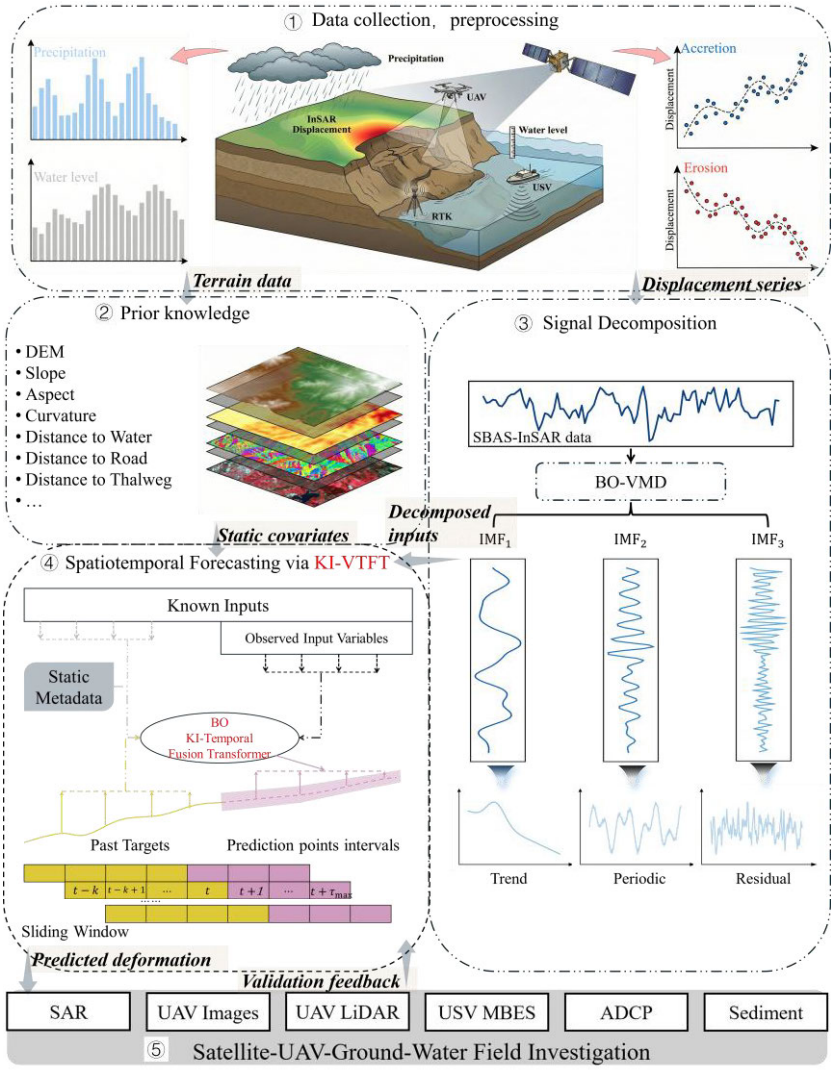


Fig. 2. Overall architecture and data flow of the KI-VTFT framework. Multisource observations acquired in Module ① feed two parallel knowledge-infusion pathways: static geomorphic attributes encoded in Module ② (the spatial-level prior) and BO-VMD signal decomposition in Module ③ (the signal-level prior), which separates the displacement series into trend, periodic, and residual components. Both pathways converge on the TFT backbone in Module ④ for multistep forecasting, and Module ⑤ closes the loop through satellite–UAV–ground–water field validation that verifies the physical plausibility of the model-identified drivers.

deformation patterns and the model-identified driving factors (Section V-B).

B. Bayesian-Optimized VMD

SBAS-InSAR-derived riverbank displacement constitutes a nonstationary time series characterized by complex trend and seasonal dynamics. Physically, the total displacement $D(t)$ can be modeled as an additive superposition of three components driven by distinct mechanisms [45]: a trend component $T(t)$ governed by cumulative erosion or aggradation (controlled by bank stratigraphy and geometry); a periodic component $P(t)$ reflecting seasonal hydro-meteorological oscillations (precipitation and water levels); and a residual component $R(t)$ encompassing high-frequency variability, including both measurement noise and transient environmental disturbances. Accordingly, the signal is formulated as

$$D(t) = T(t) + P(t) + R(t) \tag{1}$$

where $T(t)$, $P(t)$, and $R(t)$ denote the trend, periodic, and residual components, respectively.

To isolate these components, we employ VMD, an adaptive signal processing technique that partitions the input signal into a discrete set of band-limited intrinsic mode functions (IMFs) [46]. VMD solves a constrained variational optimization problem designed to minimize the aggregate bandwidth of the modes while preserving signal reconstruction fidelity

$$\begin{cases} \underset{\{u_k\},\{\omega_k\}}{min} \left\{ \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \\ \text{s.t.} \sum_{k=1}^{K} u_k(t) = D(t) \end{cases} \tag{2}$$

where $u_k(t)$ and ω_k denote the kth IMF and its center frequency, respectively; K is the number of modes; $\delta(t)$ is the Dirac distribution; and $*$ denotes the convolution operator, such that $(\delta(t) + j/\pi t) * u_k(t)$ yields the analytic signal of mode u_k. This constrained problem is solved via the alternating direction method of multipliers (ADMMs) using an augmented

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Lagrangian formulation, in which a quadratic penalty factor α controls the tradeoff between mode bandwidth and reconstruction fidelity. The values of both K and α are jointly optimized through the BO procedure described below.

The efficacy of VMD is highly sensitive to two hyperparameters: the number of modes K and the quadratic penalty factor α. Conventional empirical selection often results in mode mixing or spurious components lacking physical interpretability. To resolve this, we utilized BO to automate hyperparameter tuning. BO constructs a probabilistic surrogate model of the objective function and iteratively selects candidate parameters by maximizing an acquisition function, thereby efficiently exploring the parameter space to identify optimal configurations with minimal computational cost [47]. Crucially, the BO framework facilitates the optimization of the entire KI-VTFT pipeline, simultaneously tuning both VMD decomposition parameters and subsequent TFT model hyperparameters (Table SI). The objective optimized in this search is the multistep forecasting error of the complete pipeline, for which exhaustive grid or random search over the joint space of the VMD and TFT parameters would be computationally prohibitive. In contrast to standard VMD with an empirically fixed mode number, this data-adaptive selection of K and α aligns the recovered modes with the three-component additive model of (1). Both hyperparameters are searched over fixed ranges, with K drawn from the integer interval [2], [10] and α from [100, 3000] (Table SI). For each candidate K, the extracted IMFs are sorted by center frequency and aggregated into the three components of (1) by a deterministic rule, in which the lowest-frequency mode forms $T(t)$, the highest-frequency mode forms $R(t)$, and the remaining mid-frequency modes are summed to form $P(t)$. The aggregation is executed automatically within every BO evaluation and involves no manual interpretation of the spectra. At the selected $K = 3$, each group contains exactly one mode and the aggregation reduces to an identity mapping, a degenerate case of the same rule. The three-component representation itself is prescribed by the physical model of (1); what the search over [2], [10] determines adaptively is the mode number, whose convergence to $K = 3$ places the recovered modes in one-to-one correspondence with that representation. All three reconstructed components are supplied to the TFT encoder as independent observed input features. Retaining $R(t)$ is essential because, beyond measurement noise, this component can encapsulate transient deformation anomalies associated with episodic hydro-meteorological forcing and its delayed geotechnical response. The extent to which $R(t)$ preserves such event-driven signals is quantitatively examined in Section IV-B1.

C. Temporal Fusion Transformer

The TFT serves as the predictive backbone of the KI-VTFT framework. This architecture leverages self-attention mechanisms and recurrent sequence modeling to capture complex temporal dependencies while maintaining interpretability. As illustrated in Fig. 3, the model integrates five core components: gated residual networks, variable selection networks, static covariate encoders, interpretable multihead attention, and quantile output layers [21], [48].

1) Gated Residual Network: The GRN functions as the fundamental structural unit controlling information flow. For a primary input a and optional context vector c, the GRN computes

$$\eta_2 = \mathrm{ELU}\left(W_{2,\omega}a + W_{3,\omega}c + b_{2,\omega}\right) \tag{3}$$

$$\eta_1 = W_{1,\omega}\eta_2 + b_{1,\omega} \tag{4}$$

$$\mathrm{GRN}_\omega(a,c) = \mathrm{LayerNorm}\left(a + \mathrm{GLU}_\omega(\eta_1)\right) \tag{5}$$

where $\eta_1, \eta_2 \in R^{d_{model}}$ are intermediate layers, and ω denotes shared weights. The exponential linear unit (ELU) ensures efficient gradient flow, while the gated linear unit (GLU) dynamically modulates feature expression based on learned gating weights.

2) Variable Selection Network: This module identifies relevant inputs from the heterogeneous feature set at each time step. Let $\xi_t^{(j)}$ be the transformed input of the jth variable at time t, and let $[\Xi]_t = [\xi_t^{(1)^T}, \ldots, \xi_t^{(m_\chi)^T}]^T$ be the flattened input vector. Selection weights are computed as

$$v_{\chi t} = \mathrm{Softmax}\left(\mathrm{GRN}_{v_\chi}\left([\Xi]_t, c_s\right)\right) \tag{6}$$

where c_s is the static context vector. Each feature undergoes independent processing

$$\tilde{\xi}_t^{(j)} = \mathrm{GRN}_\xi\left(\xi_t^{(j)}\right). \tag{7}$$

The final weighted composite feature vector is obtained via

$$\tilde{\xi}_t = \textstyle\sum_{j=1}^{m_\chi} v_{\chi t}^{(j)} \tilde{\xi}_t^{(j)}. \tag{8}$$

This mechanism explicitly quantifies the contribution of each input—including BO-VMD components, precipitation, and water levels—enabling transparent attribution of deformation drivers.

3) Static Covariate Encoder: Static geomorphic attributes are processed through dedicated encoders to generate four context vectors: c_s for variable selection, c_e and c_c for LSTM state initialization, and c_h for static enrichment. These time-invariant covariates condition temporal dynamics on site-specific terrain characteristics, thereby injecting spatial susceptibility priors into the sequential learning process.

4) Interpretable Multihead Attention: A modified self-attention mechanism captures long-range dependencies [49]. Standard attention weights are computed as

$$\mathrm{Attention}(Q,K,V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d_{attn}}}\right)V \tag{9}$$

where $Q, K \in R^{N \times d_{attn}}$ and $V \in R^{N \times d_v}$. To ensure interpretability, TFT employs a multihead variant with shared value weights

$$\mathrm{InterpretableMultiHead}(Q,K,V) = \frac{1}{m_H}\sum_{h=1}^{m_H}\mathrm{Attention}\left(QW_Q^{(h)}, KW_K^{(h)}, VW_V\right)W_O \tag{10}$$

where $W_V \in R^{d_{model} \times d_v}$ is shared across all m_H heads, and W_O is the output linear transformation weight. Unlike standard multihead attention, where each head employs an independent value projection $W_v^{(h)}$ that maps inputs into head-specific subspaces, TFT shares a single W_v across all heads

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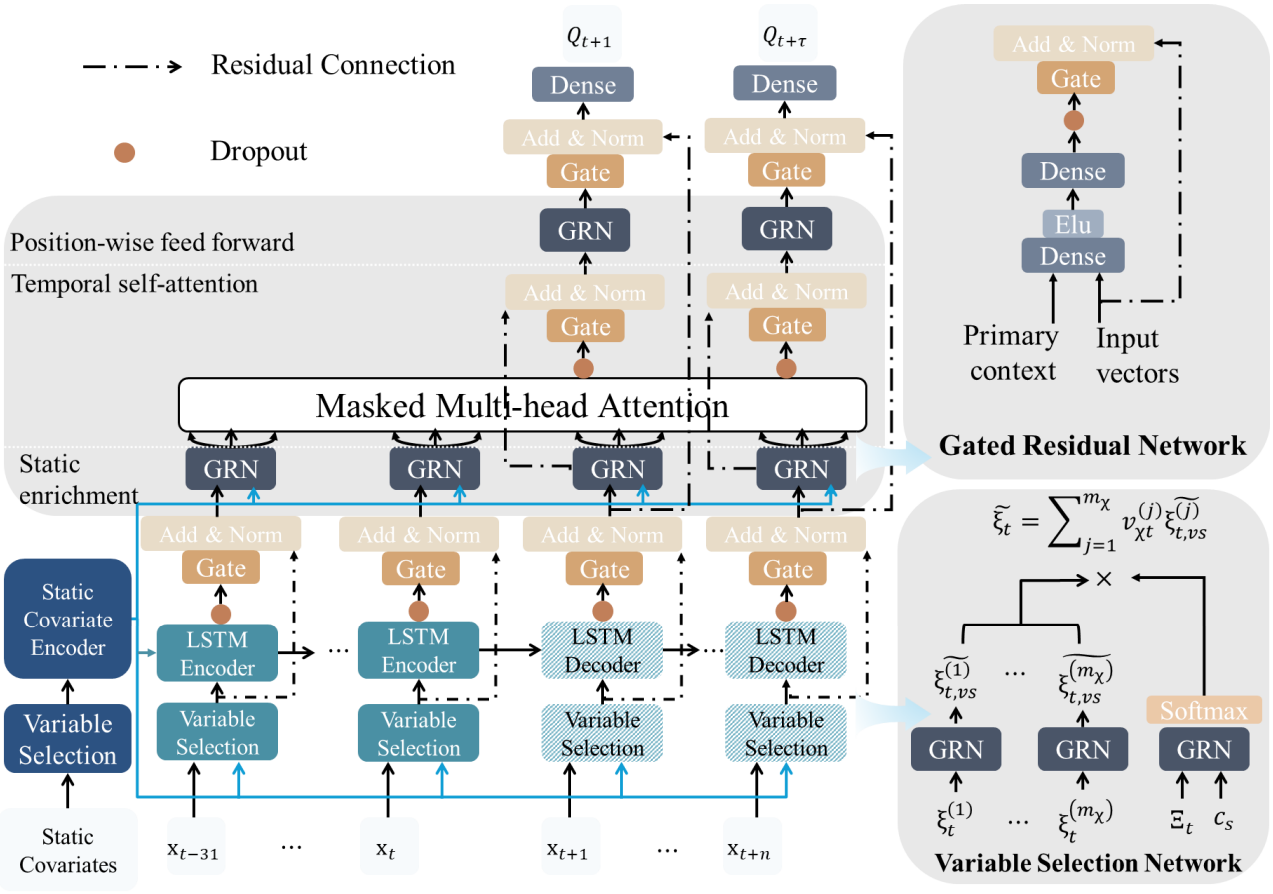


Fig. 3. Architecture of the TFT.

[see (10)]. This architectural choice has a direct consequence for interpretability: because every head attends over the same value representation, the per-head attention weights become commensurable and can be legitimately averaged into a single global temporal importance score $1/m_H \sum_{h=1}^{m_H} A^{(h)}$. The resulting score quantifies the relative degree to which each historical time step contributes to the decoder output. This structurally grounded temporal attribution is a property of the original TFT design [21] and is absent from standard Transformer architectures with head-specific value projections, which is why the TFT is adopted as the backbone here. It underpins the temporal analyses in Section V-A, where the attention weights reveal which historical periods within the 31-step look-back window, about one year, drive the forecast.

5) Quantile Forecasting: To quantify uncertainty in bank collapse prediction, the model generates probabilistic forecasts across a dense quantile set $Q = \{0.10, 0.20, 0.25, 0.50, 0.75, 0.90, 0.98\}$. The output for the qth quantile is generated via linear transformation of the temporal fusion decoder output $\tilde{\psi}(t,\tau)$:

$$\hat{y}(q,t,\tau) = W_q \tilde{\psi}(t,\tau) + b_q \tag{11}$$

where $\tau \in \{1,\ldots,\tau_{max}\}$ spans the prediction horizon. We configure $\tau_{max} = 5$ steps (corresponding to a 60-day horizon given the 12-day resolution). The network is trained by minimizing the quantile loss

$$\mathrm{L}(\Omega, W) = \sum_{y_t \in \Omega} \sum_{q \in Q} \sum_{\tau=1}^{\tau_{max}} \frac{QL\left(y_t, \hat{y}\left(q, t-\tau, \tau\right), q\right)}{M\tau_{max}} \tag{12}$$

$$QL(y,\hat{y},q) = q(y-\hat{y})_+ + (1-q)(\hat{y}-y)_+ \tag{13}$$

where $(\cdot)_+ = \max(0,\cdot)$. This objective function produces calibrated prediction intervals, providing critical information on exceedance probabilities for hazardous displacement thresholds.

The displacement time series from each monitoring point was partitioned chronologically into training and test subsets at an 8:2 ratio, preserving temporal order to prevent future information leakage. Specifically, the first 80% of the time steps were used for model training and hyperparameter optimization via BO with fivefold temporal cross-validation, while the remaining 20% served as a held-out test set for performance evaluation. All input features were standardized to zero mean and unit variance using statistics computed exclusively from the training partition. The model was trained using the Adam optimizer with the optimal hyperparameters listed in Table SI. Training was conducted on two NVIDIA A800 GPUs. The sensitivity of the forecasting performance to the VMD decomposition scope is examined in Section IV-B3.

To ensure physical credibility, only monitoring points with temporal coherence $\geq$ 0.6 and deformation

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patterns corroborated by independent field observations (Section IV-A2) were included in model training.

D. Performance Evaluation

To quantitatively evaluate the forecasting accuracy and generalization capability of the KI-VTFT framework, we employ four standard statistical metrics: RMSE, mean absolute error (MAE), Pearson correlation coefficient (CC), and the coefficient of determination (R^2) [50]. These metrics are defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i^p - y_i^t\right)^2} \tag{14}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i^p - y_i^t\right| \tag{15}$$

$$CC = \frac{\sum_{i=1}^{n}\left(y_i^p - \overline{y^p}\right)\left(y_i^t - \overline{y^t}\right)}{\sqrt{\sum_{i=1}^{n}\left(y_i^p - \overline{y^p}\right)^2}\sqrt{\sum_{i=1}^{n}\left(y_i^t - \overline{y^t}\right)^2}} \tag{16}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i^p - y_i^t\right)^2}{\sum_{i=1}^{n}\left(y_i^t - \overline{y^t}\right)^2}. \tag{17}$$

In these formulations, n denotes the number of test samples; y_i^p and y_i^t represent the predicted and observed displacement values at time step i, respectively; and $\bar{y^p}$ and $\bar{y^t}$ represent the means of the predicted and observed sequences. Lower RMSE and MAE values indicate higher predictive precision, whereas CC and R^2 values approaching unity signify strong correlation and goodness-of-fit between model outputs and ground truth.

Furthermore, to quantify the performance gain of KI-VTFT relative to baseline models, we define the improvement ratio (IR)

$$IR = \frac{S_{base} - S_{ours}}{S_{base}} \times 100. \tag{18}$$

In this metric, the error scores (RMSE or MAE) of the proposed KI-VTFT model S_{ours} are compared against those of the benchmark model S_{base}. A positive IR value explicitly indicates the percentage reduction in prediction error achieved by the proposed framework.

IV. Results

A. SBAS-InSAR Deformation and Hydrometeorological Context

1) Spatial Pattern of Bank Instability: Fig. 4 illustrates the spatial distribution of SBAS-InSAR-derived displacement velocities and their statistical characteristics for the Hechangzhou and Nanyunhe reaches (November 2021–December 2024). The deformation field exhibits marked spatial heterogeneity. In the Hechangzhou Reach, the island interior is dominated by positive velocities (2.4–14.5 mm/yr), indicative of surface accretion or seasonal rebound, whereas localized negative-velocity clusters (<−7.4 mm/yr, red pixels) clearly delineate active bank-retreat zones. Notably, two monitoring points on the northern margin of Hechangzhou reveal contrasting evolutionary modes governed by hydrodynamic zonation: H_1 registers persistent negative velocities, corresponding to active retreat driven by high shear stress proximal to the thalweg; conversely, H_2 displays positive velocities (blue), reflecting localized aggradation within a near-bank mixing zone, where reduced flow velocity promotes sediment deposition despite adjacent channel incision. Statistical analysis of the coherent pixel ensemble yields a mean velocity of $\mu = 0.65$ mm/yr and a standard deviation of $\sigma = 4.45$ mm/yr, confirming that while the majority of the surface remains stable, the distribution tails ($|v| > 7.4$ mm/yr) effectively isolate discrete instability foci [Fig. 4(b)]. In contrast, the Nanyunhe Reach features a continuous negative-velocity corridor (from −19.7 to −7.4 mm/yr) extending along the riparian strip at transects N_1 and N_2, transitioning landward to stable conditions. Field photographs at H1 [Fig. 4(c)] and H2 [Fig. 4(d)] illustrate the contrasting bank conditions at these two monitoring points.

The observed spatial velocity pattern is consistent with the geomorphic zonation inherent to bifurcated channel systems. Hechangzhou Island and the Nanyunhe are separated by the active river channel and governed by fundamentally different processes, namely, floodplain aggradation on the island interior and near-bank erosion along the Nanyunhe corridor. This bifurcated-channel pattern, in which mid-channel islands and bars undergo persistent accretion while adjacent riparian corridors experience concentrated scour, is well-documented in the long-term evolution of large alluvial rivers [3], [51], [52]. Two cross-sectional velocity profiles [AA' and BB'; Fig. 4(e) and (f)] further confirm that velocity transitions from positive to negative values coincide with geomorphic boundaries rather than occurring as spatially smooth gradients. Profile BB' additionally reveals negative velocities (from −10 to −15 mm/yr) at the northern island margin near H1, indicating that the island surface is not characterized by uniform accretion but by spatially differentiated deformation controlled by local hydrodynamic exposure.

2) InSAR Reliability Verification: Before evaluating forecasting performance, four independent lines of evidence were examined to confirm the reliability of the SBAS-InSAR deformation field.

Stable-Area Validation: Two stable areas (SA1 and SA2) on built-up surfaces along the Nanyunhe were selected to evaluate background measurement precision [Fig. 4(a)]. SA1 comprised 455 pixels with a mean coherence of 0.85 and yielded a mean velocity of −0.39 mm/yr and a standard deviation of 0.51 mm/yr. SA2 comprised 452 pixels with a mean coherence of 0.80 and yielded a mean velocity of −0.46 mm/yr and a standard deviation of 0.38 mm/yr (Table I). These submillimeter background velocities are one to two orders of magnitude smaller than the deformation signals at active monitoring points, where absolute velocities exceed 7 mm/yr, confirming that the measurement precision is sufficient to resolve the geomorphic signals of interest.

Cross-Sectional Velocity Profiles: Velocity profiles extracted along AA' and BB' [Fig. 4(e) and (f)] exhibit sharp transitions that coincide with geomorphic boundaries. A complete data gap between approximately 4000 and 5300 m corresponds to the river channel where no coherent pixels exist. This pattern is inconsistent with atmospheric artifacts, which would manifest as spatially smooth gradients unrelated to surface features.

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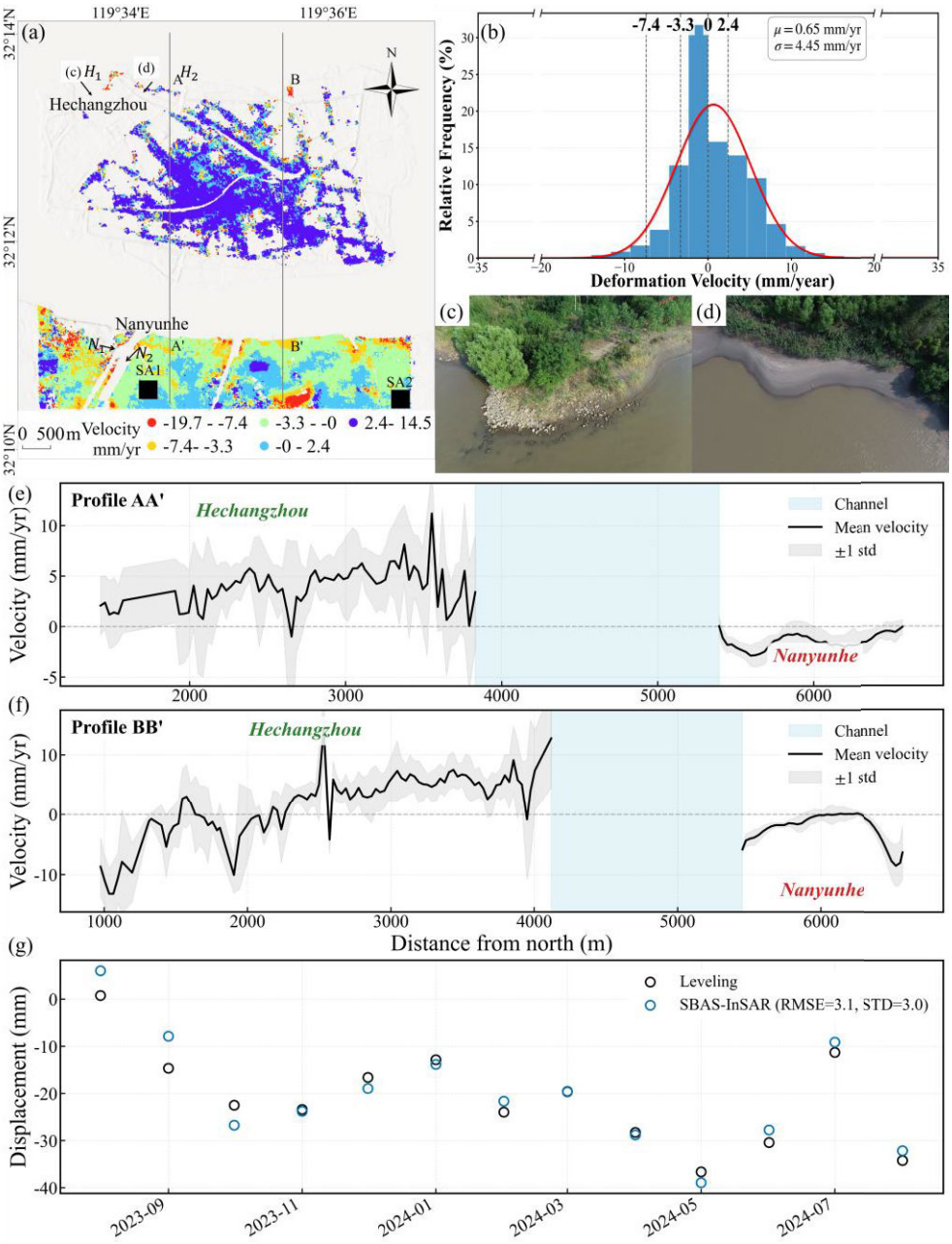


Fig. 4. Spatiotemporal characteristics of riverbank deformation in the study area. (a) SBAS-InSAR-derived velocity map for Zhenyang Reach (Hechangzhou and Nanyunhe), with cross-sectional lines AA' and BB', stable areas SA1 and SA2, and monitoring points H1, H2, N1, and N2 annotated. (b) Frequency distribution histogram of displacement velocities with a superimposed Gaussian fit. (c) and (d) UAV imagery of representative field sites. (e)–(f) Cross-sectional velocity profiles along AA' and BB', showing the spatial transition from island accretion to south-bank erosion separated by the river channel (shaded). (g) Comparison between the SBAS-InSAR and leveling displacement series.

TABLE I
STABLE-AREA VALIDATION STATISTICS

Stable Area	N pixels	Mean velocity (mm/yr)	Std (mm/yr)	Mean coherence
SA1	455	−0.39	0.51	0.85
SA2	452	−0.46	0.38	0.80

Field-Evidence Cross-Referencing: Table II summarizes the directional consistency between InSAR-derived velocities and independent field observations at all four monitoring points. Negative velocities at H1 (−18.65 mm/yr), N1 (−7.41 mm/yr), and N2 (−7.79 mm/yr) correspond to documented erosion and bed degradation, while the positive velocity at H2 (+11.18 mm/yr) corresponds to measured near-bank deposition. This systematic agreement across repeat

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TABLE II
CROSS-REFERENCE OF INSAR VELOCITIES AND INDEPENDENT FIELD OBSERVATIONS

Monitoring point	InSAR velocity (mm/yr)	Independent field evidence	Physical interpretation
H1 (Hechangzhou)	−18.65	Exposed cobble substrate, destabilized vegetation (Fig. 4c); near-thalweg scour up to 7.73 m (Fig. 11a)	Active retreat driven by thalweg-proximal high shear stress
H2 (Hechangzhou)	+11.18	Vegetated sand flat (Fig. 4d); near-bank deposition (Fig. 11a)	Aggradation in low-velocity mixing zone
N1 (Nanyunhe)	−7.41	Continuous bed degradation, max 1.89 m (Fig. 11b); near-bed velocity 26.45 cm/s (Fig. 12b); D50 = 36.8 μm (Fig. 12a)	Systemic erosion by coupled hydrodynamic forcing and low substrate resistance
N2 (Nanyunhe)	−7.79	Same erosion corridor as N1	Persistent near-bank erosion

TABLE III
SUBREACH ANNUAL DEFORMATION STATISTICS

Year	Sub-reach	Mean velocity (mm/yr)	Std (mm/yr)	Positive fraction (%)	Negative fraction (%)
2022	Hechangzhou	+7.80	9.48	83.5	16.5
2022	Nanyunhe	-2.07	5.04	28.0	72.0
2023	Hechangzhou	+1.88	7.86	65.4	34.6
2023	Nanyunhe	-2.71	4.63	17.0	83.0
2024	Hechangzhou	+4.80	9.18	76.5	23.5
2024	Nanyunhe	-1.79	6.00	27.5	72.5

UAV–USV bathymetry, ADCP flow profiling, sediment sampling, and UAV imagery provides converging evidence that the deformation field captures genuine geomorphic processes rather than processing artifacts.

Leveling Benchmarking: The InSAR vertical displacement, averaged over the coherent pixels surrounding a leveling benchmark, was further compared with the in situ precise leveling series measured there. The two series agree, after atmospheric correction, at an RMSE of 3.1 mm (standard deviation 3.0 mm), improved from 4.0 mm (3.9 mm) beforehand [Figs. 4(g) and S2(a)]. The individual error contributions are summarized in an uncertainty budget (Table SV).

3) Interannual Deformation Variability and Hydrological Forcing: To characterize the interannual variability of bank instability, we calculated annual displacement velocity statistics for Zhenyang Reach from 2022 to 2024 [Fig. 5(a)–(f)]. Given the pronounced spatial heterogeneity between Hechangzhou and Nanyunhe documented in Section IV-A1, the annual statistics were computed both at the whole-reach level and separately for each subreach (Table III). In 2022, the mean velocity was μ = 3.26 mm/yr (σ = 8.47 mm/yr), with 40.28% of coherent points falling within the moderate accretion/rebound range (2.4–14.5 mm/yr) and only 1.12% exhibiting strong negative velocities (< −19.7 mm/yr), indicating that strong erosion was spatially confined. A distinct shift occurred in 2023: the mean velocity declined to μ = −0.23 mm/yr (σ = 6.52 mm/yr), the moderate positive fraction contracted to 22.82%, and the proportion of negative-velocity points rose appreciably, signaling a spatial expansion of instability despite lower peak magnitudes. In 2024, the mean velocity recovered to μ = 1.77 mm/yr (σ = 7.87 mm/yr), with the positive class rebounding to 38.54%; however, the strong negative fraction increased to 1.44%, reflecting intensified localized erosion despite a positive whole-reach mean.

Hechangzhou maintained positive mean velocities in all three years (+7.80, +1.88, and +4.80 mm/yr for 2022, 2023, and 2024, respectively), with the positive-velocity fraction ranging from 65.4% to 83.5%, indicating persistent accretion-dominated behavior on the island surface. Nanyunhe recorded negative mean velocities in all three years (−2.07, −2.71, and −1.79 mm/yr), with the negative-velocity fraction exceeding 72.0% in each year. The whole-reach annual means reported above (μ = 3.26, −0.23, and 1.77 mm/yr) therefore reflect the combined outcome of these two contrasting geomorphic units, rather than a region-wide shift between uplift and subsidence. The near-zero whole-reach mean in 2023 (μ = −0.23 mm/yr) corresponds to a year when weakened Hechangzhou aggradation (+1.88 mm/yr) was nearly offset by Nanyunhe erosion (−2.71 mm/yr), not to a cessation of geomorphic activity.

The year-to-year modulation of these subreach rates is physically consistent with the observed hydrological variability. In 2022, the highest flood-season mean water level (4.02 m) sustained prolonged overbank conditions that promoted sediment settling on the island surface (+7.80 mm/yr), while the persistent high-stage flow intensified near-bank scour along Nanyunhe (−2.07 mm/yr). In 2023, the lowest flood-season mean water level (3.90 m) reduced overbank frequency, weakening island aggradation to +1.88 mm/yr; however, the occurrence of a higher flood peak (5.13 versus 4.64 m in 2022) maintained south-bank erosion at −2.71 mm/yr. In 2024, the largest annual precipitation (1423.50 mm) and highest flood peak (5.43 m) combined to sustain moderate island aggradation (+4.80 mm/yr) while partially replenishing the south-bank

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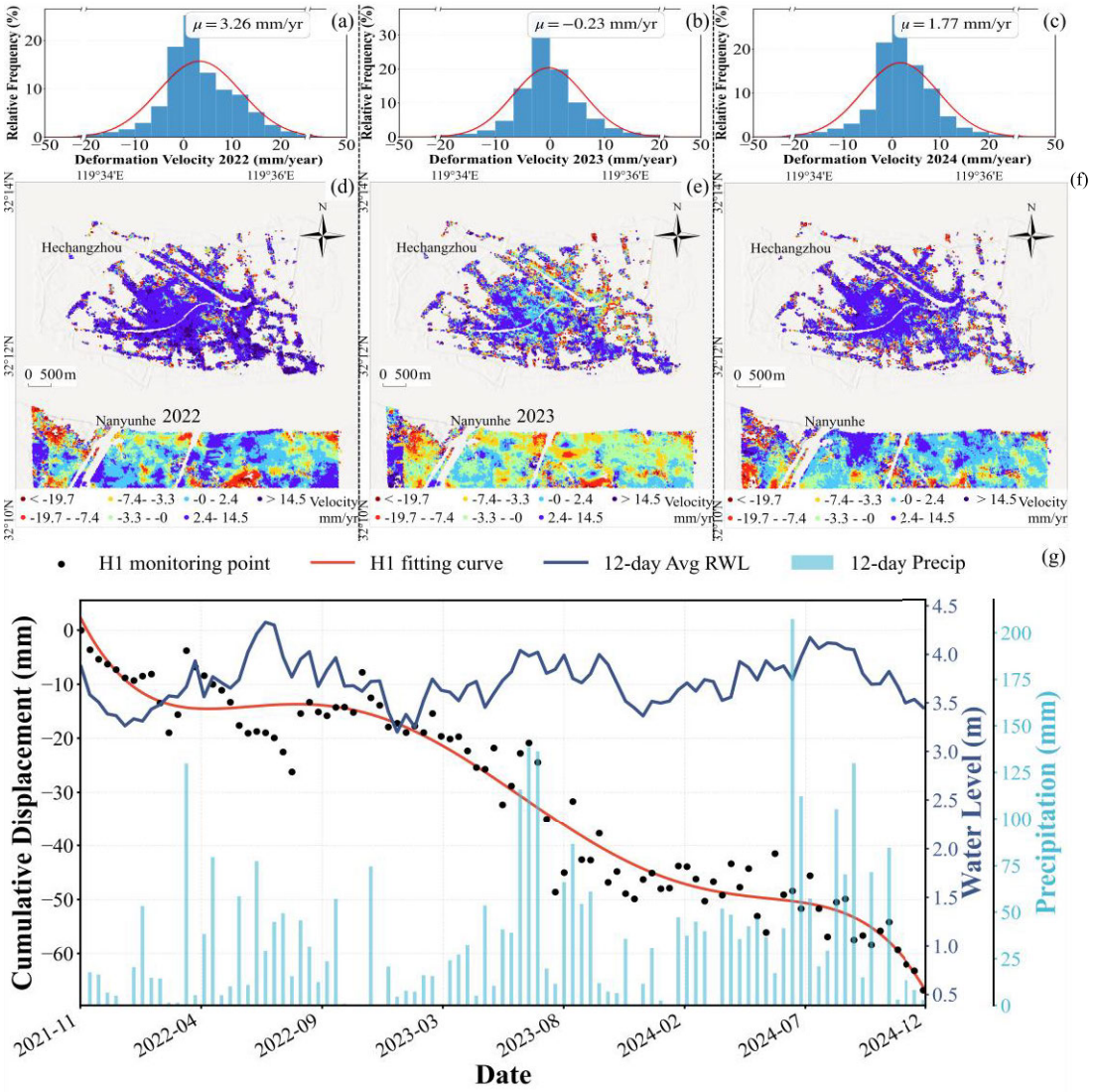


Fig. 5. (a)–(f) Comparison of annual displacement velocities across calendar years 2022–2024 for Zhenyang Reach. (d)–(f) Share a unified color scale. (g) Temporal evolution of cumulative displacement at $\boldsymbol{H}_1$ plotted against 12-day cumulative precipitation and 12-day average water level.

corridor with fine sediment, reducing Nanyunhe erosion to −1.79 mm/yr. The interannual stable-area analysis further confirms that these velocity fluctuations are geomorphic in origin rather than processing artifacts: SA1 and SA2 yielded annual velocities within mm/yr across all three years (SA1: +0.08, −0.35, and −0.31 mm/yr for 2022, 2023, and 2024; SA2: +0.18, −0.59, and −0.29 mm/yr), with no systematic year-to-year drift.

Fig. 5(g) illustrates the temporal correlation between the cumulative displacement time series at monitoring point H_1 and external drivers (12-day average water level and cumulative precipitation). Cumulative displacement showed a progressive decline from 0 to approximately −67 mm, punctuated by step-like accelerations that coincide with elevated river stages and intense precipitation pulses. The 12-day average water level fluctuated seasonally between 3.21 and 4.34 m, while precipitation peaked at ∼208 mm in early July 2024. These temporal correspondences suggest that bank deformation is governed by coupled hydro-meteorological forcing: rising river stages enhance bank-toe saturation and seepage gradients, while antecedent rainfall elevates pore-water pressure within the bank matrix, collectively reducing shear resistance and triggering episodic retreat. The spatial contrast between island aggradation and south-bank erosion held in all three years (Table III); interannual hydrological variability governed the rate of each process, not its direction.

B. Evaluation of Model Effectiveness

1) BO-VMD Signal Decomposition Results: Following the methodology detailed in Section III-B, we applied the BO-VMD framework to decompose the raw SBAS-InSAR displacement time series. The BO process converged to an optimal hyperparameter configuration of $K = 3$ and $\alpha = 400$. This configuration successfully partitioned the complex deformation signal into three physically distinct components: 1) a trend term capturing long-term cumulative deformation; 2) a periodic term reflecting recurring hydro-meteorological

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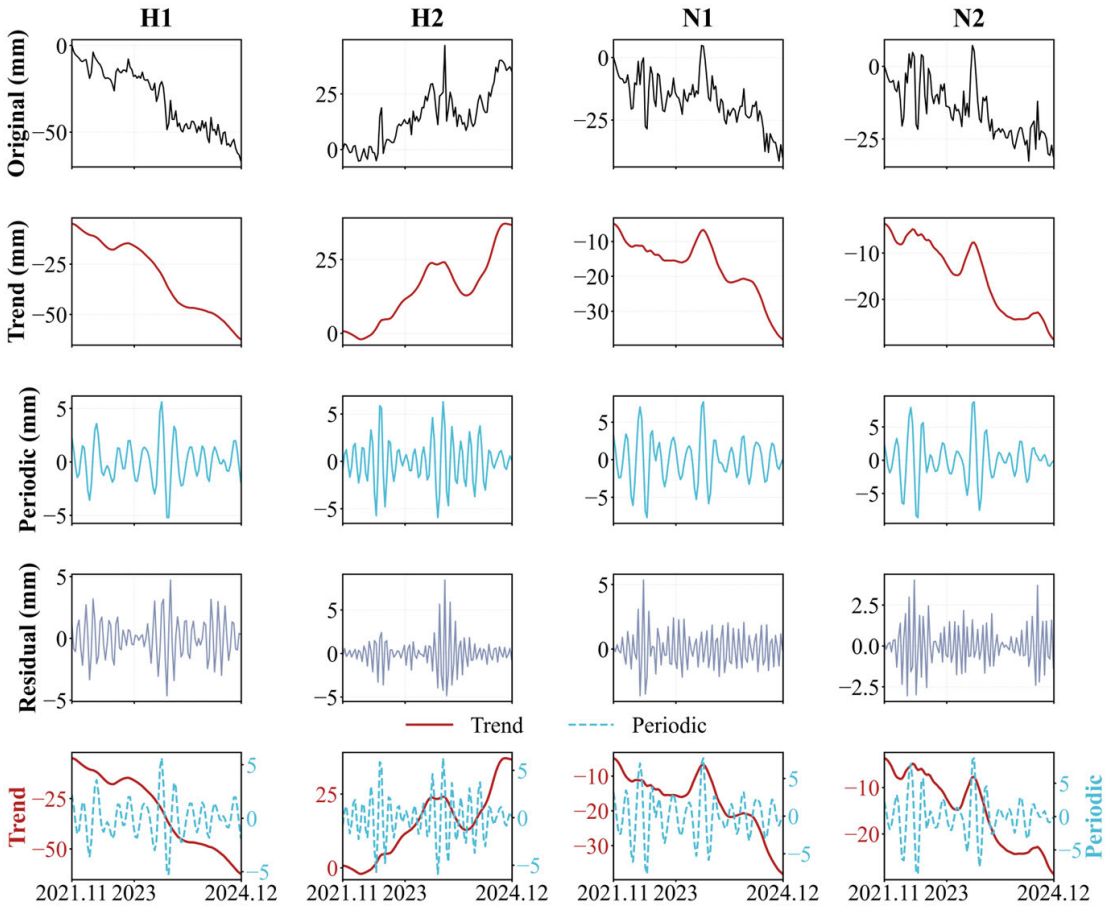


Fig. 6. BO-VMD decomposition results for monitoring points $\boldsymbol{H}_1$, $\boldsymbol{H}_2$, $\boldsymbol{N}_1$, and $\boldsymbol{N}_2$.

oscillations; and 3) a residual term representing stochastic measurement noise.

Fig. 6 presents the decomposition results for four representative monitoring points across contrasting geomorphic settings. At H_1, the trend component exhibits a monotonic decline from −4.92 to −62.38 mm (net change −57.46 mm), consistent with the sustained bank retreat documented in Section IV-A1. Conversely, H_2 displays a steadily ascending trend from 0.71 to 36.77 mm (net change +36.06 mm), plateauing near 37.20 mm in late 2024; this corroborates the localized aggradation regime within the hydrodynamic mixing zone. Both N_1 and N_2 show monotonically declining trend curves, with net changes of −33.24 and −24.96 mm, respectively, indicating persistent erosion along the Nanyunhe near-bank corridor. The periodic components exhibit amplitude ranges of 11–17 mm across all four points. Fast Fourier transform (FFT) analysis reveals dominant periods of approximately 61–106 days (~2–3 months), likely corresponding to subseasonal water-level fluctuation cycles rather than annual hydrological rhythms. Residual amplitudes range from 7 to 13 mm, accounting for measurement noise and aperiodic disturbances. This decomposition effectively partitions the complex displacement sequence into component-specific subsignals: trend and periodic capture the deterministic dynamics, while the residual retains high-frequency variability that may include both stochastic noise and event-driven transient anomalies. All three components are input to the TFT encoder, enabling componentwise forecasting that leverages long-term evolutionary trajectories, multimonth periodic patterns, and episodic displacement signals.

To examine whether the VMD framework preserves transient deformation signals, we performed a componentwise energy-partition analysis at a representative coherent point that exhibited abrupt displacement excursions during the 2024 flood season (Fig. S3). Three transient episodes were identified in July 2024 using a threshold of mean $\pm 2\sigma$ applied to the interepoch displacement increments. During these transient periods, the residual component concentrated 96.7% of the total decomposed-signal energy, with trend and periodic contributing only 0.9% and 2.4%, respectively [Fig. S3(c)]. In contrast, during quiescent intervals, the energy budget was dominated by trend (97.5%), while the residual fraction dropped to 0.4%. The energy-proportion ratio of residual between transient and quiescent periods reached 239.3, indicating that abrupt deformation anomalies are channeled predominantly into the residual mode rather than being suppressed by the band-limited trend or periodic components. Furthermore, the temporal peaks of the residual amplitude exhibited a visible correspondence with antecedent peak river stage and concentrated precipitation events [Fig. S3(d)], with the residual response appearing to follow the hydro-meteorological maxima rather than occurring

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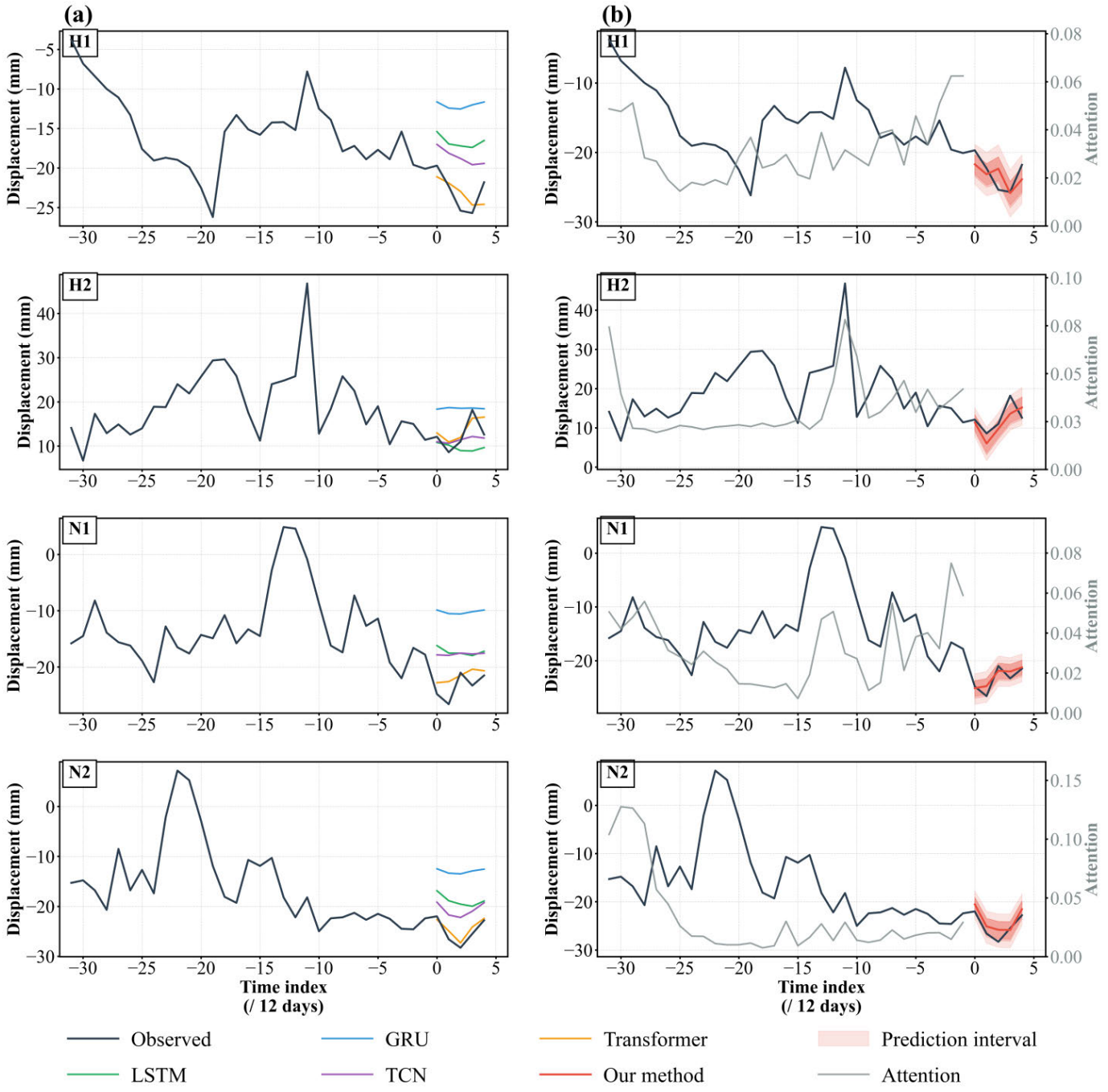


Fig. 7. Multihorizon forecasting results at representative monitoring points H_1, H_2, N_1, and N_2. (a) Comparison of observed displacement (dark blue) with baseline model predictions (LSTM, GRU, TCN, and Transformer) over the prediction horizon. (b) KI-VTFT predictions (red line) with quantile-based prediction intervals (red shading) and temporal attention weights (gray line, right axis). The predictions shown correspond to the representative run (random seed 42).

simultaneously. This temporal pattern is consistent with the delayed pore-pressure accumulation mechanism discussed in Section IV-A3, and supports the interpretation that the residual signal reflects physically meaningful transient deformation rather than purely random noise. These results suggest that the three-component VMD framework is capable of preserving episodic displacement anomalies within the residual term, which is subsequently supplied to the TFT encoder as an independent observed input.

2) Multihorizon Forecasting Performance Comparison: To evaluate the extrapolation capability of the proposed KI-VTFT framework, we compared its multihorizon forecasting accuracy against four established deep learning baselines—GRU, LSTM, TCN, and the vanilla Transformer—across one-step ($T + 1$), three-step ($T + 3$), and five-step ($T + 5$) prediction horizons at all four monitoring points (Table IV and Fig. 7). To ensure a fair comparison, all models shared an identical experimental protocol. The input feature set was

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TABLE IV
MULTIHORIZON FORECASTING PERFORMANCE COMPARISON AMONG GRU, LSTM, TCN, TRANSFORMER, TRANSFORMER_SE, AND KI-VTFT

Indicators/IR		GRU	LSTM	TCN	Transformer	Transformer_SE	KI-VTFT
1-step (T+1)	MAE	3.83±0.18 (61.96%)	2.69±0.15 (45.84%)	2.13±0.18 (31.44%)	1.84±0.11 (20.78%)	1.77±0.05 (17.43%)	1.46±0.08
	RMSE	6.37±0.14 (68.89%)	4.29±0.15 (53.88%)	3.03±0.15 (34.58%)	2.56±0.06 (22.70%)	2.42±0.13 (18.08%)	1.98±0.06
	R^2	0.768±0.013	0.899±0.009	0.950±0.005	0.963±0.003	0.967±0.004	0.978±0.002
3-step (T+3)	MAE	3.88±0.09 (58.80%)	2.94±0.09 (45.60%)	2.19±0.17 (26.93%)	1.95±0.17 (17.95%)	1.84±0.06 (12.96%)	1.60±0.05
	RMSE	6.47±0.12 (66.97%)	4.55±0.11 (53.08%)	3.23±0.18 (33.88%)	2.61±0.09 (18.03%)	2.53±0.05 (15.53%)	2.14±0.08
	R^2	0.766±0.013	0.881±0.006	0.940±0.007	0.962±0.002	0.963±0.002	0.974±0.003
5-step (T+5)	MAE	4.01±0.16 (58.79%)	3.17±0.13 (47.93%)	2.44±0.12 (32.24%)	2.19±0.07 (24.66%)	2.07±0.19 (20.17%)	1.65±0.03
	RMSE	6.57±0.21 (65.12%)	4.81±0.17 (52.45%)	3.57±0.13 (35.92%)	3.09±0.11 (25.96%)	2.78±0.14 (17.55%)	2.29±0.06
	R^2	0.761±0.016	0.863±0.011	0.927±0.005	0.945±0.004	0.957±0.004	0.970±0.001

Note: Mean ± standard deviation over five random seeds.

unified across all architectures: each model received the three VMD-decomposed components (trend, periodic, and residual) together with the five hydro-meteorological dynamic covariates as time-varying inputs, and the seven static geomorphic attributes described in Section II-B4. For the baseline models (GRU, LSTM, TCN, and Transformer), which lack a dedicated static covariate encoder, the static attributes were repeated along the temporal axis and concatenated with the dynamic features at each time step. To verify that this repeat-and-concatenate treatment does not place the strongest baseline at a disadvantage, we additionally evaluated Transformer_SE, a vanilla Transformer equipped with a static encoder that maps the seven attributes into a context vector added to the input embeddings, while omitting the variable selection networks, gating layers, and quantile heads of the TFT, with the optimized hyperparameters of Transformer retained. For KI-VTFT, the same attributes were injected through the TFT static covariate encoder (Section III-C3). All models used the same chronological 80%/20% train–test partition, z-score normalization computed exclusively from training-set statistics, and an identical BO procedure (Optuna TPE sampler with equivalent trial budgets and random seeds). Each architecture was subsequently trained under five random seeds (42, 0, 1, 2, and 3) with the optimized hyperparameters held fixed; reported metrics are five-seed means ± standard deviations unless noted otherwise. The optimized hyperparameters and model complexity metrics for all six architectures are summarized in Table SII. While all models achieved acceptable performance in single-step forecasting, their accuracy diverged substantially as the prediction horizon extended. GRU and LSTM exhibited the steepest degradation: at $T + 5$, RMSE reached 6.57 ± 0.21 mm and 4.81 ± 0.17 mm, respectively, with corresponding R^2 values declining to 0.76 and 0.86. TCN and Transformer demonstrated improved robustness yet still incurred RMSE values of 3.57 ± 0.13 mm and 3.09 ± 0.11 mm at the longest horizon; the static-encoder variant Transformer_SE reached 2.78 ± 0.14 mm. In contrast, KI-VTFT maintained the lowest error across all horizons, achieving RMSE values of 1.98 ± 0.06, 2.14 ± 0.08, and 2.29 ± 0.06 mm for $T + 1$, $T + 3$, and $T + 5$, respectively, with R^2 remaining at or above 0.97. The IR over the vanilla Transformer reached 22.70%, 18.03%, and 25.96% in RMSE across the three horizons, and 17.55% over Transformer_SE at the five-step horizon, confirming that the knowledge-guided architecture sustains predictive stability where purely data-driven models degrade.

These margins are stable across training seeds: relative to Transformer, KI-VTFT lowered the pooled five-step RMSE by 0.80 ± 0.06 mm (paired t, df = 4, $p < 0.001$; exact one-sided Wilcoxon, $p = 0.031$; improvement in five of five seeds), corresponding to a seed-level IR of 25.9% ± 1.3% (95% confidence interval, 24.3%–27.6%), and relative to Transformer_SE by 0.49 ± 0.16 mm (paired t, df = 4, $p = 0.001$; Wilcoxon, $p = 0.031$; five of five seeds). Equipping the Transformer with an equivalent static encoder closes 0.31 mm of this 0.80 mm margin; the remaining 0.49 mm reflects the knowledge-infused architecture itself rather than the handling of static features. Differences between successive models in Table IV exceed the pooled five-seed standard deviation by factors of 2.5–9.1.

Beyond accuracy, we further examined whether the performance gain of KI-VTFT is accompanied by a justifiable cost in model complexity. As summarized in Table SII, KI-VTFT contains 1.16 M trainable parameters, which is of the same order of magnitude as the Transformer baseline (0.93 M) and less than 1.25 times larger. Test-set inference time per full test pass was 4.97 s for KI-VTFT, compared with 1.03 s for Transformer, 2.08 s for TCN, 1.61 s for LSTM, and 0.67 s for GRU; all values remain at the subsecond-per-sample level

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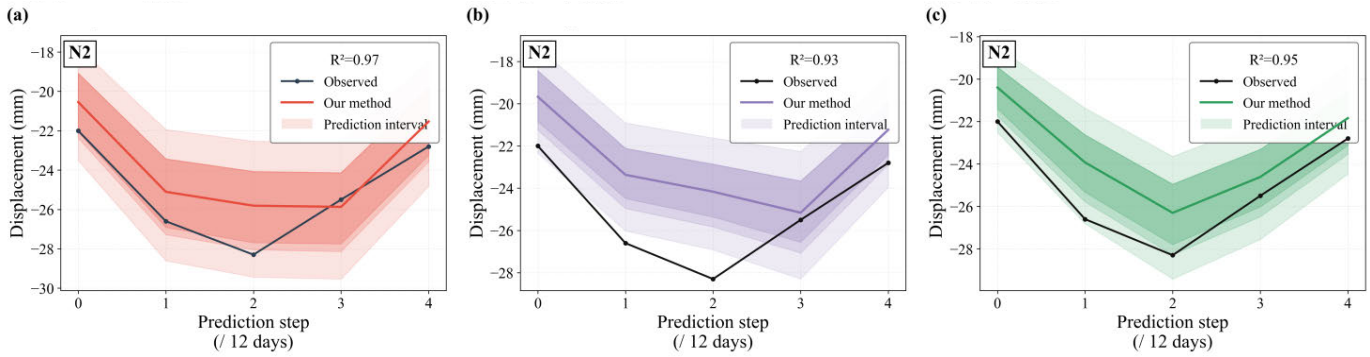


Fig. 8. (a) Full-sequence VMD (red), where decomposition is applied to the entire time series before the train–test split (R^2 = 0.97). (b) Train-only VMD (purple), where decomposition is fit exclusively on the training partition and applied to the test sequence without refitting (R^2 = 0.93). (c) Rolling VMD (green), where the decomposition is updated by incrementally incorporating new observations (R^2 = 0.95). Shaded regions denote quantile-based prediction intervals. The three panels show the representative run (random seed 42).

and are compatible with operational 12-day Sentinel-1 revisit cycles. The modest additional cost of KI-VTFT is attributable to the three interpretability-oriented modules inherited from the TFT backbone (variable selection networks, static covariate encoder, and interpretable multihead attention). These modules not only yield the 25.96% RMSE reduction over Transformer at the five-step horizon (Table IV), but also provide the feature and temporal attribution discussed in Section V-A and the zero-shot transferability demonstrated in Section V-C, capabilities that the Transformer baseline does not offer. The advantage of KI-VTFT is therefore not limited to a marginal accuracy gain but extends to interpretability and cross-site generalization at a comparable model scale.

3) Assessment of VMD Decomposition Strategy: In the primary experiments reported above, VMD decomposition was applied to the full displacement time series prior to the chronological train–test split. This full-sequence protocol is consistent with prevailing practice in recent decomposition-based geohazard prediction studies [16], [24], [25], [26] and facilitates direct comparison with these benchmarks. However, because the decomposition basis functions are informed by future observations, this protocol may introduce a degree of optimistic bias in the reported metrics.

To quantify this effect, we conducted a comparative experiment using a train-only VMD protocol. In this setting, VMD was fit exclusively on the training partition (the first 80% of the time series), and the resulting decomposition parameters were applied to the held-out test sequence without refitting. Fig. 8 presents the five-step forecasting results at monitoring point N_2 under the three strategies. The full-sequence setting [Fig. 8(a)] achieved R^2 = 0.97, whereas the train-only setting [Fig. 8(b)] yielded R^2 = 0.93. The decline of 0.04 in R^2 confirms that full-sequence decomposition introduces a modest optimistic bias. Nevertheless, the train-only configuration still demonstrates strong predictive capability, with R^2 exceeding 0.90 and the predicted trajectory faithfully reproducing the observed deformation trough.

This observation is consistent with the findings of Zhou et al. [25], who reported that restricting VMD to training data produced slightly larger errors while maintaining comparable overall accuracy in landslide displacement prediction. The limited performance gap indicates that the predictive advantage of the KI-VTFT framework is predominantly attributable to the structured integration of signal-level and spatial-level priors within the TFT architecture. The full-sequence results reported in Table IV and Fig. 7 therefore remain representative of the framework's forecasting capability, while the train-only results provide a complementary conservative reference suitable for operational early warning deployment where strictly causal data processing is required. A rolling VMD scheme that incrementally incorporates new observations [Fig. 8(c)] yielded R^2 = 0.95, confirming that such strictly causal, real-time processing retains the framework's accuracy. This online updating removes the need for a complete precollected record at a new site, and the trained network applies to an analogous reach without retraining (Section V-C), together defining a causal deployment workflow.

C. Contribution Analysis of Key Components

To quantify the specific contribution of each module within the KI-VTFT framework, we conducted a systematic analysis by sequentially isolating three core components: 1) Bayesian hyperparameter optimization (denoted w/o optimization; abbreviated as BO in Taylor diagrams); 2) BO-VMD signal decomposition (denoted w/o BO-VMD; abbreviated as VM); and 3) static geomorphic attributes (denoted w/o static; abbreviated as ST) (Table SIII). The TFT backbone served as the invariant predictive engine and was retained across all variants. Taylor diagrams (Fig. 9) visualize the comparative performance of these variants against the full KI-VTFT model (abbreviated KI) across the four monitoring points. The IRs reported in this section are computed from the four-point mean given in the overall row of Table SIII, where the corresponding per-point values are also tabulated.

Experimental results indicate that model performance deteriorates significantly upon the removal of any single component, confirming the indispensability of each module. The most pronounced degradation occurred when BO was omitted. Relative to the BO variant, the complete KI-VTFT achieved an IR of 55.7% in RMSE and 48.3% in MAE, with R^2 improving from 0.87 to 0.97. Because the optimization step

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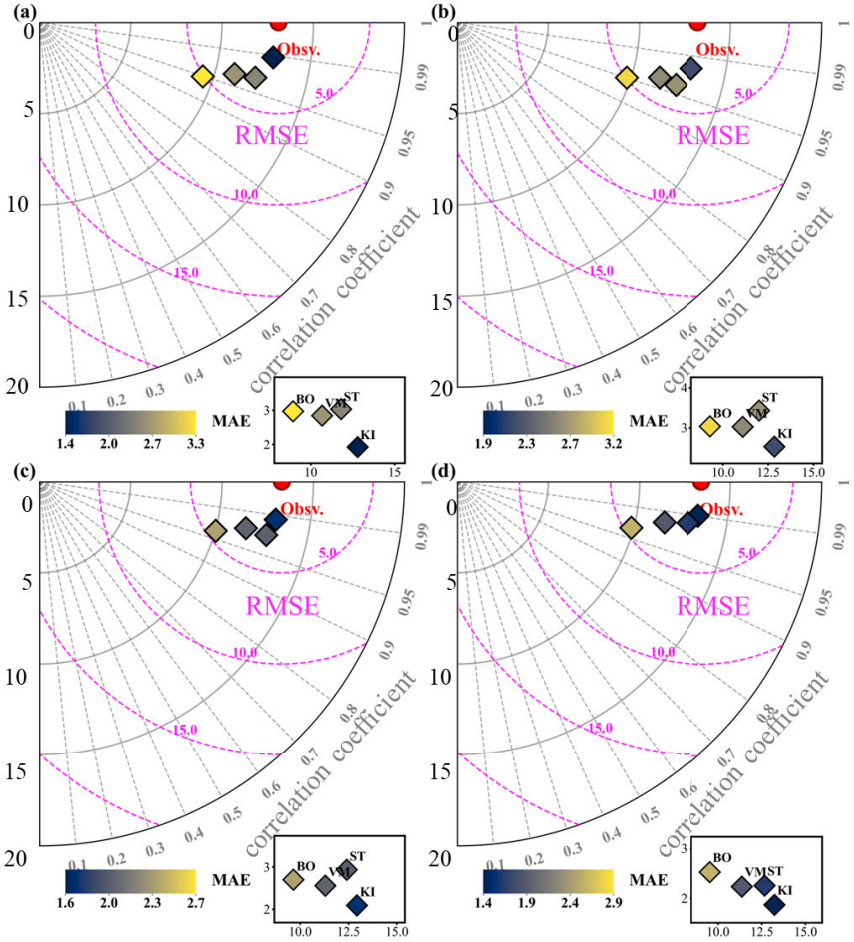


Fig. 9. Taylor diagrams comparing the statistical performance of the complete KI-VTFT model (KI) against partial variants (BO: w/o optimization; VM: w/o BO-VMD; and ST: w/o static) at four monitoring points: (a) H_1, (b) H_2, (c) N_1, and (d) N_2. Statistics are computed from the representative run (random seed 42).

tunes both the VMD decomposition and the TFT backbone, we next isolate the contribution of each.

A two-by-two factorial experiment independently switched BO of the VMD decomposition on or off (BO-VMD versus standard VMD) and hyperparameter tuning of the TFT backbone on or off (tuned versus default), with all four configurations evaluated over five seeds (Fig. S4). With both stages optimized, the framework reached an RMSE of 2.13 mm; replacing BO-VMD with standard VMD raised the error to 3.01 mm, reverting the TFT to default settings raised it to 2.43 mm, and reverting both reproduced the w/o optimization variant at 4.81 mm. Removing the VMD optimization therefore costs 0.88 mm against 0.30 mm for the TFT tuning, and this 0.88 mm increase is significant across seeds (paired t-test, $p < 0.001$, with all five seeds agreeing in sign). The two factors also interacted significantly (1.51 mm, $p < 0.001$); the contribution of each optimization was larger when the other was withheld. The larger contribution comes from optimizing the VMD decomposition, the step that builds the signal-level prior. The exclusion of the BO-VMD module yielded the second-largest error increase. The KI-VTFT model attained an IR of 39.0% in RMSE and 33.1% in MAE over the VM variant, with R^2 rising from 0.93 to 0.97. These findings confirm that inputting raw displacement sequences directly into the TFT, without prior decomposition into trend and periodic subsignals, substantially compromises accuracy, underscoring the value of signal-level priors in disentangling deterministic dynamics from stochastic noise.

The ST variant, which excludes the seven geomorphic attributes, exhibited the smallest yet notable performance gap. The full model achieved an IR of 31.8% in RMSE and 26.9% in MAE relative to this variant, with R^2 improving from 0.94 to 0.97. Although less severe than the losses induced by removing optimization or decomposition, this degradation confirms that static terrain encoding provides essential spatial context that enhances forecasting precision, particularly in geomorphically complex bank segments.

The Taylor diagrams (Fig. 9) provide a visual synthesis of these findings. Across all monitoring points, the complete KI-VTFT model (KI) clusters closest to the observational reference (Obsv.), exhibiting the highest CC and lowest centered root-mean-square difference. Conversely, the BO variant deviates most substantially, falling near the 5-mm RMSE contour. The VM and ST variants occupy intermediate positions, with their relative distances corresponding to the magnitude of the calculated IR values. Collectively, these results validate the synergistic contribution of BO, VMD-based decomposition, and static geomorphic encoding,

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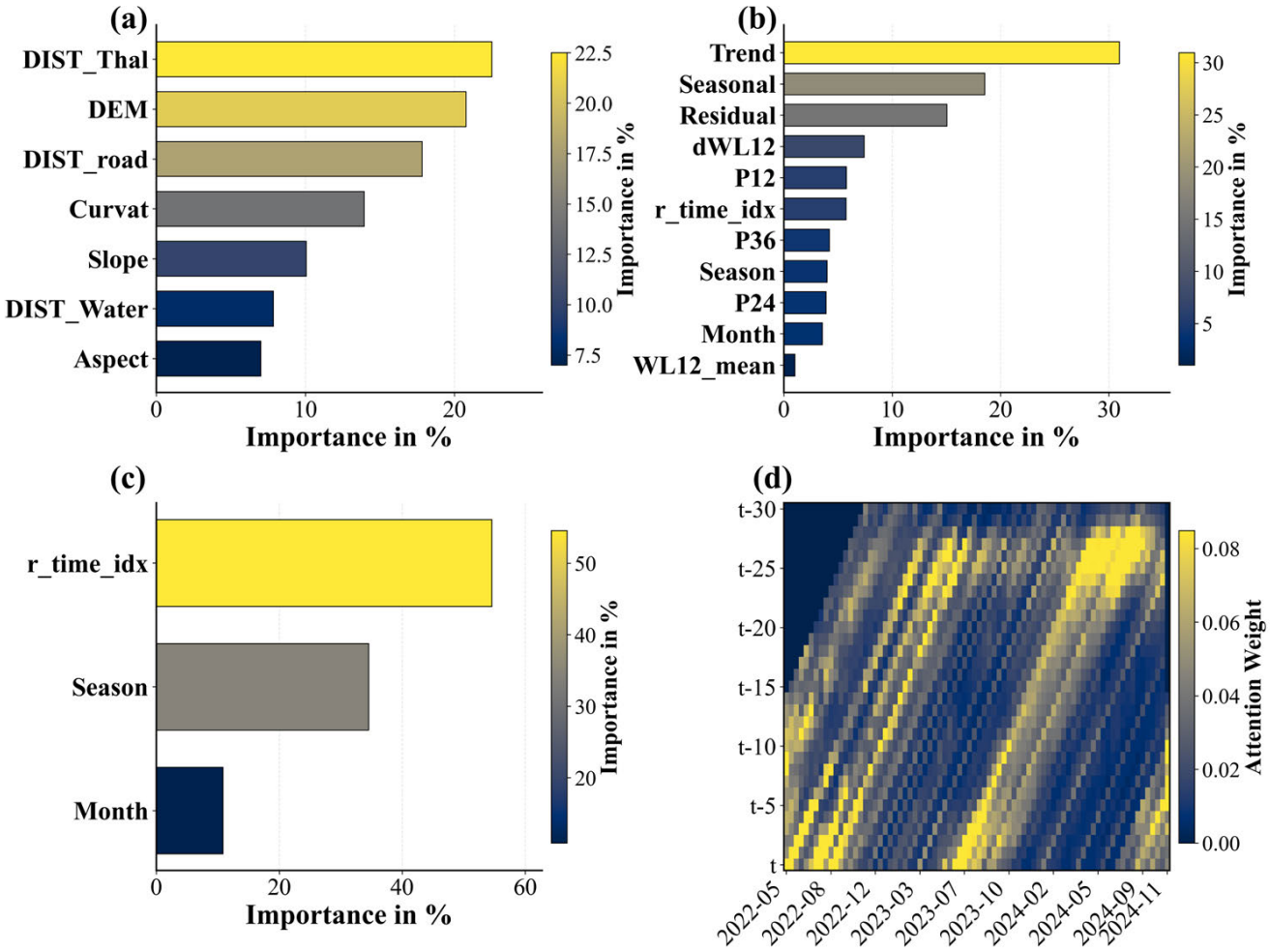


Fig. 10. Interpretability outputs of the KI-VTFT framework. (a) Static variable importance from the variable selection network. (b) Dynamic variable importance in the encoder. (c) Known future input importance in the decoder. (d) Temporal attention heatmap showing the importance weights assigned to the 31 historical time steps.

confirming that the knowledge-guided design philosophy of injecting both signal-level and spatial-level priors is essential for robust riverbank deformation forecasting. Beyond the removal of individual components, the choice of BO-VMD as the decomposition front-end was assessed by benchmarking it against five alternative schemes under an otherwise identical pipeline, namely, standard VMD, empirical mode decomposition (EMD), complete ensemble EMD with adaptive noise (CEEMDAN), singular spectrum analysis (SSA), and a no-decomposition control (Table SIV). At the five-step horizon, BO-VMD attained the lowest error (RMSE 2.13 mm, R^2 0.974), improving on standard VMD by 0.88 mm, a 29% reduction, and thereby isolating the contribution of optimizing the mode number and the bandwidth penalty. SSA reached comparable accuracy (RMSE 2.19 mm) but required a 12-D embedding window (L = 12), against the three physically interpretable modes recovered by BO-VMD. Every scheme except EMD improved on the no-decomposition control; EMD performed worst and fell below that control (RMSE 3.76 versus 3.50 mm), showing that an unsuitable decomposition can impair rather than improve forecasting performance.

V. Discussion

A. Model Interpretability and Driving Factor Analysis

While accurate displacement forecasting is central to risk assessment, quantitative attribution of the underlying driving mechanisms is equally critical for effective mitigation. Unlike opaque black-box deep learning models, the KI-VTFT architecture provides interpretability through three channels inherited from its TFT backbone [21]: 1) feature importance ranking via the variable selection network; 2) temporal attention weighting via the multihead attention mechanism; and 3) seasonal dependency analysis through known future inputs. Fig. 10 visualizes these interpretability outputs, enabling a data-driven examination of the physical factors governing riverbank deformation.

Static Variable Importance: The analysis of static geomorphic attributes [Fig. 10(a)] reveals that distance to thalweg emerges as the dominant driver (22.51%). This aligns with hydrodynamic theory, where bank segments proximal to the thalweg experience intensified scour due to concentrated high-velocity flow [53]. Elevation ranks second (20.78%), reflecting the gravitational potential energy available for mass

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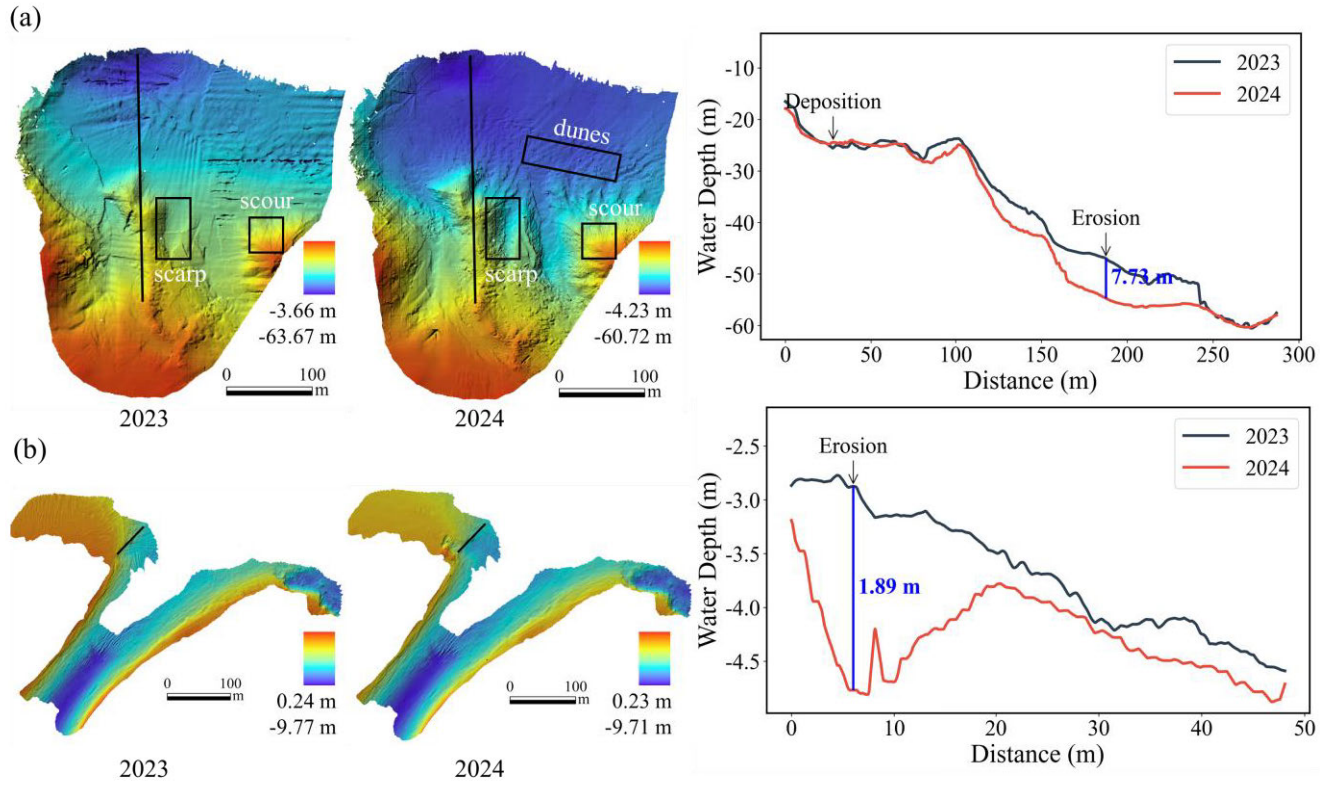


Fig. 11. UAV–USV land–water comparison between August 2023 and August 2024. (a) Hechangzhou Reach, highlighting near-bank deposition and thalweg scour. (b) Nanyunhe Reach, showing continuous bed degradation.

movement, followed by distance to road (17.84%), a proxy for anthropogenic loading. Notably, curvature (13.95%) effectively differentiates convex (erosion-prone) from concave (deposition-prone) geometries. Furthermore, slope (10.07%) and distance to water (7.85%) also play significant roles in modulating local stability. This hierarchy corroborates the SHAP-based global importance analysis from our prior susceptibility study [54], providing cross-method validation of the geomorphic controls on bank instability.

Dynamic Variable Importance: The encoder input analysis [Fig. 10(b)] demonstrates a clear dominance of VMD-decomposed signal components: trend (30.96%), periodic (18.52%), and residual (15.04%) collectively account for 64.52% of the total importance. This underscores the efficacy of signal-level knowledge priors in disentangling deterministic dynamics from stochastic noise. Regarding hydro-meteorological drivers, 12-day water-level fluctuation (dWL12, 7.39%) and 12-day cumulative precipitation (P12, 5.76%) are significant contributors, confirming that short-term precipitation and stage oscillations jointly govern deformation via coupled infiltration and seepage processes. Conversely, the 12-day mean water level (WL12_mean) ranks lowest (1.01%), implying that the rate and amplitude of stage fluctuation exert greater influence on bank stability than absolute stage height alone, a finding consistent with transient seepage mechanics [44].

Known Future Input Importance: For decoder inputs [Fig. 10(c)], relative_time_idx dominates (54.58%), followed by season (34.56%) and month (10.87%). The prominence of time indexing reflects the model's reliance on positional encoding for trajectory extrapolation, while the substantial contribution of season confirms that the TFT has successfully internalized the seasonal hydro-meteorological rhythm governing bank deformation.

Temporal Attention Patterns: The attention heatmap [Fig. 10(d)] visualizes the temporal weights assigned to historical states (y-axis: from lags t-31 to t-5) across the prediction window (x-axis: May 2022–December 2024). The heatmap exhibits pronounced diagonal striping: significant attention weights concentrate during the flood season (May–August), coinciding with peak water levels and rainfall, while dry-season intervals (October–March) show reduced weighting. Notably, attention is distributed across multiple lags rather than focusing solely on recent steps, with peaks around t-5 to t-10 and t-25. This suggests the model captures multi-scale dependencies, integrating both immediate triggers and medium-term antecedent conditions. These results confirm that riverbank deformation is governed by complex temporal associations rather than simple regression, suggesting that the model's temporal focus is consistent with known hydrological forcing patterns and may inform the identification of critical monitoring windows.

Spatial Stability of Attention Patterns: To assess whether the temporal attention patterns reflect a systematic physical signal rather than point-specific overfitting, we conducted a spatial stability analysis across 20 monitoring points spanning all

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five deformation categories (strong erosion, moderate erosion, stable, moderate deposition, and strong deposition). For each point, we computed the flood-season attention ratio, defined as the mean attention weight during May–August divided by the annual mean. All 20 points yielded ratios exceeding 1.0 (overall mean 1.66 $\pm$ 0.06; minimum 1.57), indicating that the model consistently assigns elevated attention to flood-season historical states regardless of deformation magnitude or direction [Fig. S5(c)]. The attention heatmaps for five representative points covering the full deformation spectrum exhibited visually consistent seasonal banding [Fig. S5(a)] and the lag-wise attention profiles showed narrow cross-point variability [Fig. S5(b)]. Pairwise Pearson correlations of coarse lag signatures yielded a mean of 0.92 [minimum 0.72; Fig. S5(d)], confirming that the macroscopic temporal attention structure is spatially robust.

Interpretability Scope: The interpretability afforded by the TFT architecture operates at the level of structured attribution. The variable selection network quantifies the relative contribution of each input feature through learned softmax weights, while the shared-value-weight attention mechanism provides a global temporal importance score for historical time steps (Section III-C4). These outputs identify which features and when in the historical record the model relies upon for forecasting—a capability that recent studies have similarly leveraged for geohazard applications [22], [25], [30]. Importantly, the interpretability of KI-VTFT in this study does not rest on attention weights alone, but on four mutually consistent lines of evidence. First, the variable selection network ranks the static geomorphic drivers, and this ranking independently reproduces the SHAP-based attribution from our susceptibility study [54], so two distinct attribution methods agree on the same geomorphic controls. Second, the encoder importance is dominated by the physically defined VMD components (64.52% in aggregate). Third, the flood-season concentration of the temporal attention is not inferred from a single point but is shown to be spatially stable across twenty monitoring points spanning all deformation categories (Fig. S5). Fourth, and most directly, the multiplatform field campaign in Section V-B provides external corroboration, as UAV–USV bathymetry, ADCP profiling, and sediment sampling independently confirm the hydrodynamic and geomorphic drivers identified by the model using observations entirely external to the network. Because these model-internal attributions and independent field measurements converge on the same physical controls, the evidence for physical consistency does not depend on any single interpretability channel. We characterize the KI-VTFT's interpretability as offering physically consistent empirical attributions that can inform monitoring prioritization, complementing rather than replacing process-based mechanistic analysis.

B. Physical Verification Loop: Validating AI With Field Observations

The independent multiplatform field observations served to both verify the physical credibility of the SBAS-InSAR deformation field, complementing the quantitative validation in Section IV-A2, and provide mechanistic insight into the model-identified drivers. Repeat integrated UAV–USV bathymetric surveys were conducted at the Hechangzhou and Nanyunhe reaches in August 2023 and August 2024. Two representative cross sections traversing the subaqueous slopes adjacent to the key monitoring zones were selected for detailed morphological analysis (Fig. 11).

Hechangzhou Reach: The bathymetric profile proximal to monitoring point H_2 [Fig. 11(a)] reveals distinct morphological zonation. In the near-bank zone, significant sediment deposition is evident. This accumulation is hydraulically controlled by the low-velocity mixing zone, where reduced transport capacity promotes the settling of fine-grained sediment [55]. Crucially, this subaqueous depositional signature aligns perfectly with the positive displacement velocities (aggradation) detected by SBAS-InSAR at the adjacent onshore point H_2 (Section IV-A1), suggesting that sedimentation processes operate continuously across the water-land interface. Conversely, the near-thalweg zone exhibits pronounced incision, with a maximum scour depth of 7.73-m recorded over the one-year interval. This intense erosion is attributed to high shear stress exerted by thalweg-concentrated flow. Furthermore, the channel bed develops symmetric bedforms, scarps, and scour holes, indicating active morphodynamic adjustments driven by complex recirculating flow.

Nanyunhe Reach: In contrast, the transect proximal to monitoring point N_1 [Fig. 11(b)] exhibits continuous bed degradation across the entire profile, with a maximum scour depth of 1.89 m. Unlike the Hechangzhou section, no significant depositional features are identified. This absence of accretion corroborates the persistent negative-velocity corridor detected by InSAR along the Nanyunhe riparian strip (Section IV-A1), confirming a regime of systemic erosion.

These in situ bathymetric observations provide independent physical verification of the InSAR-derived deformation patterns. The spatial correspondence between subaqueous near-bank aggradation and onshore uplift signals at Hechangzhou, juxtaposed with the parallel erosion patterns at Nanyunhe, confirms that the KI-VTFT framework captures physically meaningful morphodynamics rather than spurious correlations. This convergence of remote sensing inference and field evidence effectively closes the validation loop between AI-based prediction and process-based understanding.

Hydrodynamic and sedimentological surveys at the Nanyunhe Reach provide mechanistic insights into the observed bank instability. ADCP profiling adjacent to monitoring point N_1 reveals that flow velocity increases progressively toward the main channel, peaking in the thalweg region [Fig. 12(b)]. High velocities persist within the initial 50-m segment aligned with the main flow, while the subsequent 50–150-m transect exhibits gradual velocity attenuation. Spatially, this high-energy hydrodynamic zone aligns precisely with the continuous erosion corridor identified in the bathymetric analysis [Fig. 12(b)]. Three-dimensional vector analysis [Fig. 12(c)] indicates a robust vertical flow structure, with mean velocities of 34.26 cm/s at the surface, 27.95 cm/s at mid-depth ($0.6H$), and 26.45 cm/s at the near-bed layer.

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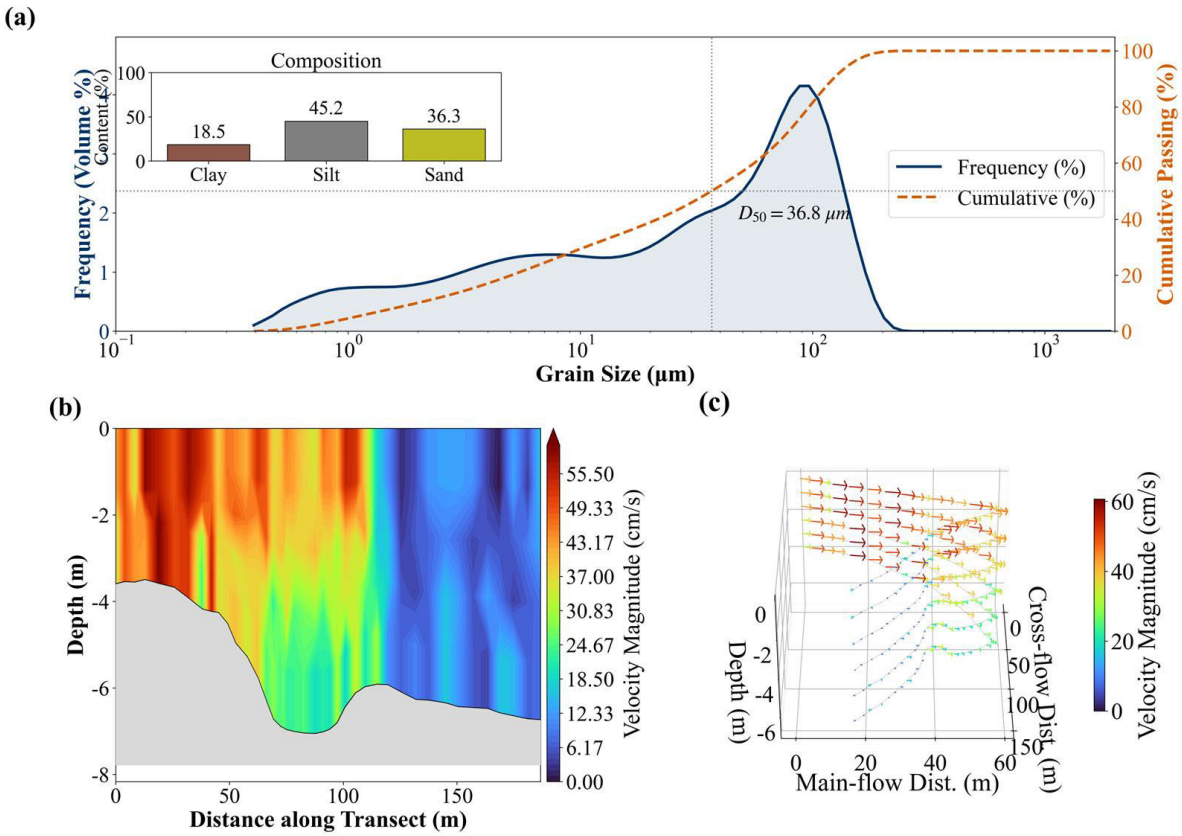


Fig. 12. Hydrodynamic and sediment characteristics at the Nanyunhe Reach adjacent to N1. (a) Grain size distribution curve and composition of bed sediment. (b) Cross-sectional flow velocity magnitude contour. (c) 3-D velocity vector field showing vertical flow structure.

This vertical distribution demonstrates that significant near-bed velocity is sustained to directly drive bed scour.

Granulometric analysis of riverbed surface samples reveals a noncohesive composition dominated by silt (45.2%) and sand (36.3%), with a median grain size (D_{50}) of 36.8 μm [Fig. 12(a)]. This fine-grained, poorly consolidated substrate exhibits low critical shear stress for incipient motion, rendering the channel highly susceptible to entrainment under the observed hydrodynamic forcing [43]. The coupling of sustained flow velocities and highly erodible sediment provides a robust physical explanation for the persistent negative displacement detected by InSAR at N_1. This confirms that the erosion patterns captured by the KI-VTFT framework are grounded in measurable physical processes rather than numerical artifacts.

C. Spatial Transferability to an Independent Reach

To rigorously assess the spatial generalizability of the proposed KI-VTFT framework, we conducted a zero-shot transfer experiment. The model, originally trained on Zhenyang Reach (Hechangzhou and Nanyunhe), was applied directly to Wuwei Reach without any fine-tuning or retraining. Although Wuwei Reach (Anhui segment, see Section II-A) shares analogous hydro-geomorphic settings with the training domain, its displacement data remained strictly unseen during model development. This cross-site validation strategy rigorously tests whether the knowledge priors embedded in KI-VTFT (specifically VMD-based decomposition and static geomorphic encoding) capture transferable physical laws rather than merely fitting site-specific artifacts. Whereas the knowledge-infused landslide model of Ma and Mei [30] encodes spatial heterogeneity through point-specific identifiers, the attribute-based conditioning adopted here allows the trained network to be applied directly to a reach absent from training.

Fig. 13 illustrates the forecasting results at a representative erosion-prone monitoring point within Wuwei Reach. The trajectories predicted by KI-VTFT exhibit high fidelity to the observed InSAR displacements, accurately reproducing both the long-term erosional trend and the superimposed seasonal fluctuations. The narrow prediction intervals (shaded regions) indicate that model confidence remains high even in this previously unseen reach. Conversely, baseline models display marked deviations from observed trajectories, particularly during rapid displacement shifts, highlighting their limited capacity for cross-site generalization.

Quantitatively, the proposed framework demonstrated consistent accuracy across the entire Wuwei Reach. For multistep prediction horizons (T + 1 to T + 5), the KI-VTFT model achieved global R^2 values ranging from 0.97 to 0.98, with MAE values of 1.61–1.99 mm and RMSE values of 2.26–2.77 mm. Even at the challenging five-step horizon, the model maintained an RMSE of 2.77 mm, consistently outperforming the strongest baseline, the Transformer (RMSE = 3.65 mm). These metrics are comparable to the

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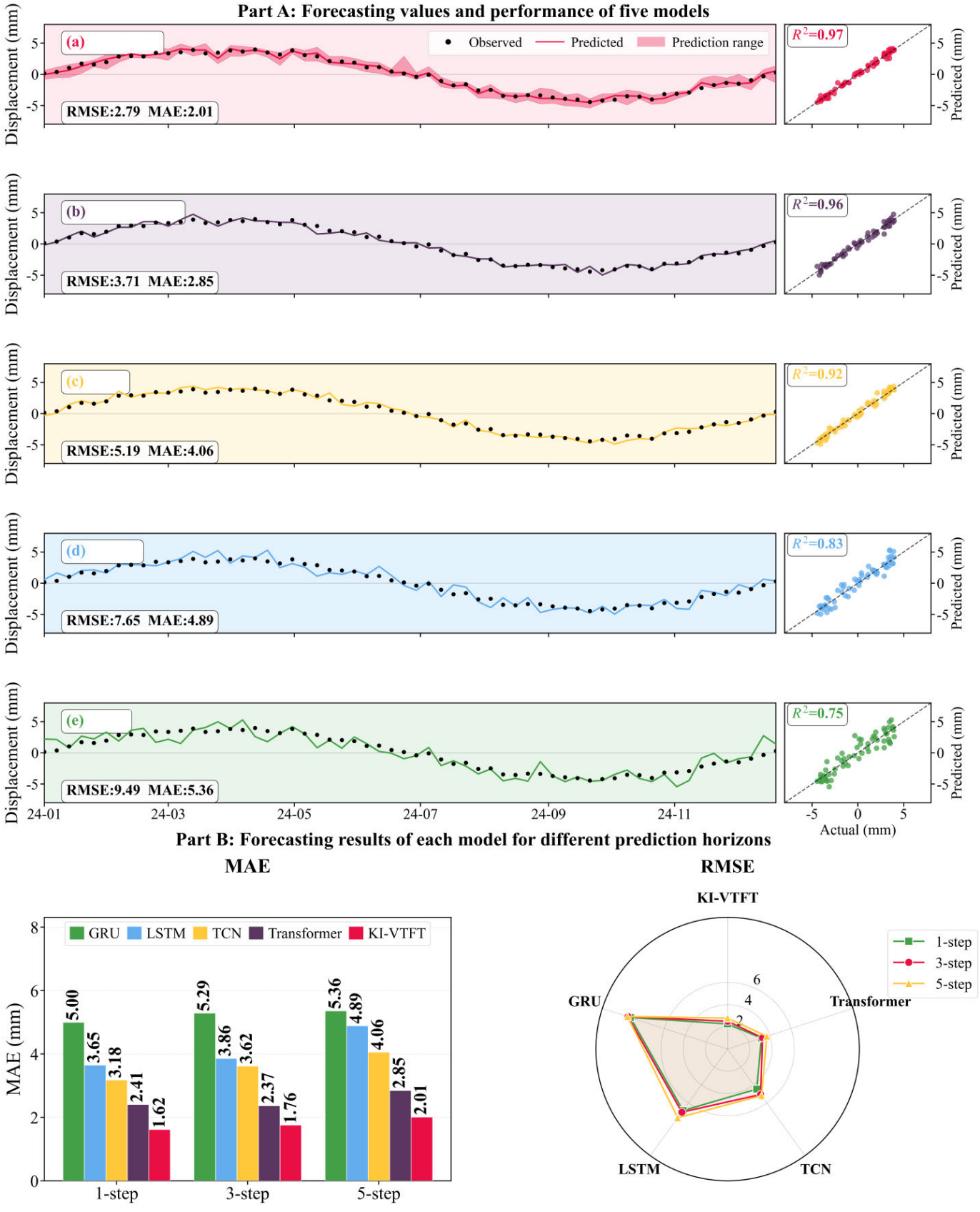


Fig. 13. Zero-shot forecasting results in Wuwei Reach. Panel metrics correspond to the representative run (random seed 42). (a) KI-VTFT. (b) Transformer. (c) TCN. (d) LSTM. (e) GRU.

performance achieved in the training domain (Section IV-B2), confirming that the model maintains predictive accuracy when transferred to a separate reach held out from training. This transfer follows from the knowledge-guided design rather than

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dataset similarity alone. The VMD front end isolates trend and periodic components whose forcing, namely, seasonal stage and rainfall, is shared by the two reaches, while the static encoder conditions on terrain attributes rather than point identifiers. The two attributes that dominate the static contribution at Zhenyang, distance to thalweg and elevation [Fig. 10(a)], are general controls on hydrodynamic exposure in meandering alluvial channels, so the attribute-to-susceptibility mapping learned on one reach carries over to another of the same fluvial style.

These findings show that KI-VTFT generalizes to an unseen reach without site-specific retraining, the property that makes direct deployment for operational early warning practical.

D. Limitations and Broader Applicability

Two limitations of this study warrant acknowledgment, concerning the signal decomposition protocol and the SAR data source. Regarding the signal decomposition protocol, the train-only VMD experiment in Section IV-B3 demonstrated that the R^2 reduction relative to full-sequence decomposition is limited to 0.04 at N_2. Regarding the SAR data source, the SBAS-InSAR processing exclusively utilized C-band Sentinel-1 imagery, which is inherently susceptible to temporal decorrelation in densely vegetated riparian zones. Future research could investigate the integration of longer wavelength L-band SAR data, such as ALOS-2 and NISAR, together with combined ascending and descending acquisitions to resolve horizontal motion components, to enhance coherence retention and deformation detection accuracy in such challenging environments [56]. Because the two sensor classes are complementary, with C-band providing the open 12-day sampling used here and L-band the penetration needed under dense vegetation, a combined acquisition strategy is the practical route for reaches with heavier canopy.

Regarding broader applicability, the proposed framework is intrinsically suited for predicting progressive riverbank instability, leveraging the sensitivity of SBAS-InSAR to slow, cumulative deformation. Bank failure typically evolves through a continuum of gradual creep, nonlinear acceleration, and ultimate collapse. In this context, the KI-VTFT model serves as a baseline for stability; if the observed displacement deviates significantly from the forecast trajectory (indicating an acceleration phase beyond the model's trend learning), this discrepancy signals a potential transition toward failure, triggering early warning protocols. Given the global coverage of open-source Sentinel-1 imagery and the zero-shot transfer shown in Section V-C, the framework can extend to comparable reaches without per-site retraining. Wuwei and Zhenyang are both meandering reaches of the lower Yangtze, so this evaluation establishes near-neighbor transfer within one fluvial system rather than cross-domain generalization. The controls underlying the transfer are specific to alluvial channels where bank retreat is scour-driven, so a markedly different geological setting, such as a bedrock-confined river whose banks are governed by rock-mass strength, would call for fine-tuning on local records.

VI. Conclusion

This study introduced KI-VTFT, a physically constrained deep learning framework that systematically embeds signal-level and spatial-level domain knowledge into a TFT to forecast riverbank deformation from Sentinel-1 InSAR time series, addressing the long-standing disconnect between data-driven prediction and process-based understanding of fluvial instability.

1) Four principal findings emerge from this work. First, KI-VTFT achieved an RMSE of 2.29 mm and R^2 of 0.97 at the five-step (60-day) horizon, reducing prediction error by 25.96%–65.12% relative to four established baselines. Ablation experiments revealed that BO, VMD decomposition, and static geomorphic encoding contributed complementarily, with their removal degrading RMSE by 55.7%, 39.0%, and 31.8%, respectively. These results demonstrate that injecting structured physical priors, rather than simply increasing model complexity, yields the most substantial accuracy gains for nonstationary fluvial deformation series. A train-only VMD experiment (Section IV-B3) yielded $R^2 = 0.93$ at the same horizon, indicating that the predictive advantage stems from the knowledge-infused architecture rather than decomposition leakage.
2) The interpretability inherited from the TFT backbone yielded attributions consistent with known physical processes. VMD-decomposed components drove 64.52% of encoder feature importance, confirming the value of explicit signal decomposition. Among static controls, distance to thalweg (22.51%) and elevation (20.78%) emerged as dominant predictors, consistent with hydrodynamic scour theory. Notably, the 12-day water-level fluctuation amplitude far exceeded mean water level in importance (7.39% versus 1.01%), indicating that the rate and magnitude of stage oscillation, rather than absolute river height, govern short-term bank deformation. This finding provides actionable guidance for monitoring agencies: early warning thresholds should prioritize stage variability metrics over static water-level benchmarks.
3) An independent validation loop integrating repeat UAV–USV bathymetry, ADCP flow profiling, and sediment sampling corroborated the InSAR-derived deformation patterns and the model-identified physical drivers. The spatial correspondence between subaqueous morphological change and onshore displacement signals closes a critical validation gap that has limited the operational credibility of deep-learning-based geohazard forecasting.
4) Zero-shot transfer to the separate Wuwei Reach maintained an RMSE of 2.77 mm and R^2 of 0.97 without retraining. This result indicates that conditioning on transferable terrain attributes rather than point-specific identifiers lets the learned susceptibility mapping carry over to a reach of similar character without recalibration.

By coupling interpretable attention mechanisms with physically grounded signal decomposition and terrain encoding,

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KI-VTFT provides a transferable basis for riverbank monitoring and early warning. Built on open-access Sentinel-1 archives, it can support monitoring in reaches where dense ground instrumentation is infeasible.

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